Numerical Valuation of Perpetual American Put Options via LCP

Project 2 (PDE BVP) Report

Team number: Group L
Yuhan Cheng Student ID: 2685836
Yijian Huang Student ID: 2685589
Department of Applied Mathematics

November 8, 2026

Abstract

This report treats the perpetual American put as an elliptic obstacle problem whose solution must satisfy a linear complementarity problem (LCP). The early exercise boundary is not prescribed in advance; it is recovered as the interface between the exercise and continuation regions. The spatial operator is discretised on a uniform grid by exponential fitting in the Scharfetter–Gummel form. The fitted weights are positive, the two off-diagonal entries are non-positive, and the row sum is strictly positive, so the discrete operator is an M-matrix at every cell Péclet number, including the degenerate region near $S = 0$. This structure yields a discrete maximum principle and a unique obstacle solution. The LCP is solved by policy iteration with sparse direct linear solves. For constant volatility and $q = 0$, the production $N = 2400$ contact bracket is $[71, 72]$ and contains the exact free boundary $S^* = 71.4286$. Because a production contact bracket is phase sensitive, the stencil order is measured separately on a boundary-aligned validation grid: the errors range from 1.073×10^{-2} to 1.682×10^{-4}, with successive ratios 3.990, 4.001, and 3.997. For local volatility and $q = 0.02$, an independent DOP853 shooting solution of the free-boundary ODE gives $S^* = 69.1664$ and $V(100) = 16.2055$. The $N = 2400$ production estimate $S_h^* = 69.1688$ lies in $[69, 70]$, whereas the coarse $N = 1200$ contact bracket $[70, 72]$ misses the reference; this coarse-grid localisation error is reported rather than treated as convergence evidence. The aligned finite-difference sequence has ratios 3.980, 4.000, and 3.999. At $S = 100$ the pure finite-domain truncation is 5.578×10^{-3}, the $N = 2400$ production total error is 4.055×10^{-3} after partial error cancellation, and the far-edge truncation is 4.833×10^{-1}. At the benchmark mesh, policy iteration requires 301 iterations and 0.045 s, whereas projected successive over-relaxation needs 96,600 iterations and 20.635 s. The measured 4.6×10^2 wall-time ratio, together with the 321-fold iteration reduction, places the acceleration in the several-hundred-fold range. On a nonuniform grid the same fitted construction reduces the maximum value error by a factor of 2.255 at $N = 2400$ when nodes are clustered around the free boundary, while an asymptotic Robin closure reduces the far-field error by up to three orders of magnitude. A discrete adjoint formulation provides the sensitivities of $V(100)$ to σ, r, and q with one extra linear solve, matching direct finite differences to 2×10^{-9} and exact derivatives to 0.31%. A controlled two-dimensional spread-put extension with $\rho = 0$ retains the five-point M-matrix structure; its pricing-window error decreases from 4.070×10^{-2} to 1.611×10^{-2} between $N = 80$ and $N = 160$, and its value at $(100, 50)$ changes by only 9.71×10^{-4} when the domain is enlarged from $L = 300$ to $L = 400$. The remaining limitations are the fully front-fitted treatment of the free-boundary kink, the finite-domain truncation of the production Dirichlet solve, and the restriction of the 2D experiment to zero correlation with approximate high-price closures.

Keywords: perpetual American put, linear complementarity problem, exponential fitting, Scharfetter–Gummel, policy iteration, free boundary, local volatility, adaptive mesh, Robin boundary, adjoint sensitivity, M-matrix

Contents

1 Introduction and Motivation **3**
- 1.1 Problem Background . . . 3
- 1.2 Mathematical Model . . . 3
- 1.3 Free-Boundary Characterisation and Analytic Benchmark . . . 4
- 1.4 Overview of the Report . . . 5

2 Discretisation and Analysis **6**
- 2.1 Discrete Scheme . . . 6
- 2.2 Consistency, Stability, and Error Structure . . . 8
- 2.3 Boundary Treatment . . . 8
- 2.4 Discrete Structure . . . 9
- 2.5 Adaptive Grids, Robin Closure, and Sensitivities . . . 11

3 Implementation **13**
- 3.1 Algorithm . . . 13
- 3.2 Data Structures and Numerical Details . . . 13
- 3.3 Computational Cost . . . 15

4 Verification and Numerical Experiments **17**
- 4.1 Convergence . . . 17
- 4.2 Adaptive Refinement and Far-Field Closure . . . 18
- 4.3 Error Separation . . . 20
- 4.4 Local Volatility . . . 22
  - 4.4.1 Four-Mesh Convergence Against the ODE Reference . . . 22
  - 4.4.2 LCP and Shape Diagnostics . . . 23
- 4.5 Main Results . . . 25
- 4.6 Adjoint Sensitivity . . . 26
- 4.7 Limitations and Robustness . . . 26
- 4.8 Boundary-Row Stress Test: Zero Flux Versus Dirichlet . . . 27
- 4.9 2D Exploratory Extension: American Spread Put . . . 28

5 Conclusions **31**

References **33**

A AI Transparency Log **34**

B Individual Contribution Statements **36**

C Supplementary Figures and Data **38**

1 Introduction and Motivation

1.1 Problem Background

An American put gives its holder the right, but not the obligation, to sell one unit of an underlying asset at strike K at any time before expiry. The value is therefore an optimal stopping value: at every state the holder compares the payoff obtained by immediate exercise with the expected discounted value obtained by continuing to hold the contract. The standard risk-neutral treatment of this decision is developed in [1, 2]. In the perpetual limit the remaining maturity tends to infinity, the time derivative disappears, and the valuation reduces to an elliptic boundary-value problem posed on the positive half-line.

The elliptic reduction retains the central difficulty of the American-option problem: the set on which exercise is optimal is not known a priori. It is separated from the continuation set by a free boundary S^*. On the exercise side the value equals the put payoff; on the continuation side it satisfies the homogeneous Black–Scholes equation; at the interface the two branches meet with continuous value and derivative. A method that solves only the partial differential equation must first locate S^* and then impose the matching conditions. An obstacle formulation avoids this two-stage procedure. It enforces the differential inequality and the payoff constraint simultaneously, and it treats the exercise region as part of the unknown solution.

This formulation is attractive for three reasons. First, the shape of the domain is fixed, so no moving-mesh or front-fixing transformation is required. Second, the problem becomes a finite-dimensional linear complementarity problem (LCP) after discretisation. Third, the discrete LCP inherits the monotonicity of the differential operator when the stencil is chosen correctly, which gives both a comparison principle and a stable active-set algorithm. The finite-difference and obstacle formulations used here follow the classical treatment in [2]; recent high-order and LCP methods for free-boundary options are represented by [3, 9].

The numerical difficulty near $S = 0$ deserves separate emphasis. The diffusion coefficient in the Black–Scholes operator is proportional to S^2, while the drift coefficient is proportional to S. Their ratio, expressed as a cell Péclet number, does not vanish as the mesh is refined. A central difference of the drift term can consequently produce a positive off-diagonal entry and destroy the M-matrix property. The resulting oscillation is not a cosmetic artefact: it violates the discrete maximum principle and can contaminate the computed exercise boundary.

Exponential fitting removes this failure by adapting the difference weights to the local ratio of drift to diffusion.

1.2 Mathematical Model

Let $S \in [0, S_{\max}]$ denote the stock price, $r > 0$ the risk-free rate, $q \geq 0$ the dividend yield, and $\sigma(S) > 0$ the local volatility. The put payoff is

$$\psi(S) = \max(K - S, 0). \tag{1}$$

Under the risk-neutral measure the stock evolves as

$$\mathrm{d}S_t = (r - q)S_t \,\mathrm{d}t + \sigma(S_t)S_t \,\mathrm{d}W_t, \tag{2}$$

and the infinitesimal generator of the discounted value is

$$\mathcal{L}V = \frac{1}{2}\sigma(S)^2 S^2 V'' + (r - q)SV' - rV. \tag{3}$$

For the perpetual put, the value is independent of calendar time and satisfies the complementarity conditions

$$V \geq \psi, \qquad -\mathcal{L}V \geq 0, \qquad (V - \psi)(-\mathcal{L}V) = 0, \tag{4}$$

or, in compact form,

$$\min\left(-\mathcal{L}V,\ V-\psi\right)=0, \qquad 0<S<S_{\max}. \tag{5}$$

The boundary conditions are

$$V(0)=K, \qquad V(S_{\max})=0. \tag{6}$$

The left condition is exact: at zero the put is certain to be exercised. The right condition is a truncation of the positive half-line and is therefore a numerical boundary condition. Its effect is quantified separately in Section 4.4.

Two parameter sets are used. The constant-volatility verification problem has

$$K=100, \qquad r=0.05, \qquad q=0, \qquad \sigma=0.20, \qquad S_{\max}=2400. \tag{7}$$

The local-volatility experiment keeps K, r, and $S_{\max}$ fixed, sets $q=0.02$, and uses

$$\sigma(S)=0.20\left[1+0.35\tanh\left(\frac{\log(S/K)}{0.40}\right)\right]. \tag{8}$$

This profile is smooth and bounded away from zero. It is below 0.20 for $S<K$ and above 0.20 for $S>K$, so it changes both the early-exercise premium and the far-field decay.

1.3 Free-Boundary Characterisation and Analytic Benchmark

For a perpetual put the exercise region is an interval $[0,S^*]$ and the continuation region is (S^*,∞). On the exercise side, $V(S)=K-S$ and hence

$$-\mathcal{L}\psi=rK-qS. \tag{9}$$

On the continuation side, V solves $-\mathcal{L}V=0$. At the free boundary the value and its first derivative are continuous:

$$V(S^*)=K-S^*, \qquad V'(S^*)=-1. \tag{10}$$

These two conditions replace an unknown boundary location by a local matching condition and are the classical form of the smooth-pasting principle.

For constant coefficients, substitution of $V(S)=S^{\lambda}$ into $-\mathcal{L}V=0$ gives the indicial equation

$$\frac{1}{2}\sigma^2\lambda(\lambda-1)+\left[(r-q)-\frac{1}{2}\sigma^2\right]\lambda-r=0. \tag{11}$$

Its roots are

$$\lambda_{\pm}=\frac{1}{2}-\frac{r-q}{\sigma^2}\pm\sqrt{\left(\frac{r-q}{\sigma^2}-\frac{1}{2}\right)^2+\frac{2r}{\sigma^2}}. \tag{12}$$

The decaying branch is λ_-; the other root grows and is inadmissible for a put. Smooth pasting gives

$$S^*=\frac{\lambda_-K}{\lambda_--1}. \tag{13}$$

For (7), $\lambda_-=-2.5$ and

$$S^*=71.4285714286, \qquad V(S)=(K-S^*)\left(\frac{S}{S^*}\right)^{-2.5}, \quad S>S^*. \tag{14}$$

In particular, $V(100)=12.3200328678$. This closed form is used only for verification; the production solver does not impose the analytic tail at the truncation boundary.

The dividend yield moves the boundary in the opposite direction to the local-volatility level. With $q=0.02$ and constant $\sigma=0.20$, the same formula gives $S^*=64.9219$. Around the local-volatility boundary the profile (8) is approximately 0.149, which lowers the value of continuation and pushes the boundary upward relative to the constant $q=0.02$ case. The computed value $S_h^*=69.17$ is the net result of these two effects.

1.4 Overview of the Report

Section 2 constructs the exponentially fitted operator, proves its M-matrix property, analyses consistency and stability, and treats the Dirichlet and free-boundary conditions; it also develops a nonuniform fitted operator, an asymptotic Robin closure, multi-point boundary extraction, and discrete adjoint sensitivities. Section 3 describes the policy-iteration algorithm and its implementation. Section 4 contains the convergence study, the two-solver comparison and a tolerance sweep that separate algebraic from discretisation error, the local-volatility and domain-truncation experiments, the adaptive-grid and Robin-closure studies, the Greeks, adjoint sensitivities, and the limitations of the approach. Section 4.9 adds a controlled two-dimensional spread-put LCP as a scaling and structure test. Section 5 summarises the conclusions. Appendices A and B contain the AI transparency log and the individual contribution statements.

2 Discretisation and Analysis

2.1 Discrete Scheme

The computational domain is divided by the uniform grid

$$S_i = ih, \qquad h = \frac{S_{\max}}{N}, \qquad i = 0, 1, \ldots, N. \tag{15}$$

At the interior nodes define

$$a_i = \frac{1}{2}\sigma(S_i)^2 S_i^2, \qquad b_i = (r-q)S_i, \qquad \mathrm{Pe}_i = \frac{b_i h}{a_i}. \tag{16}$$

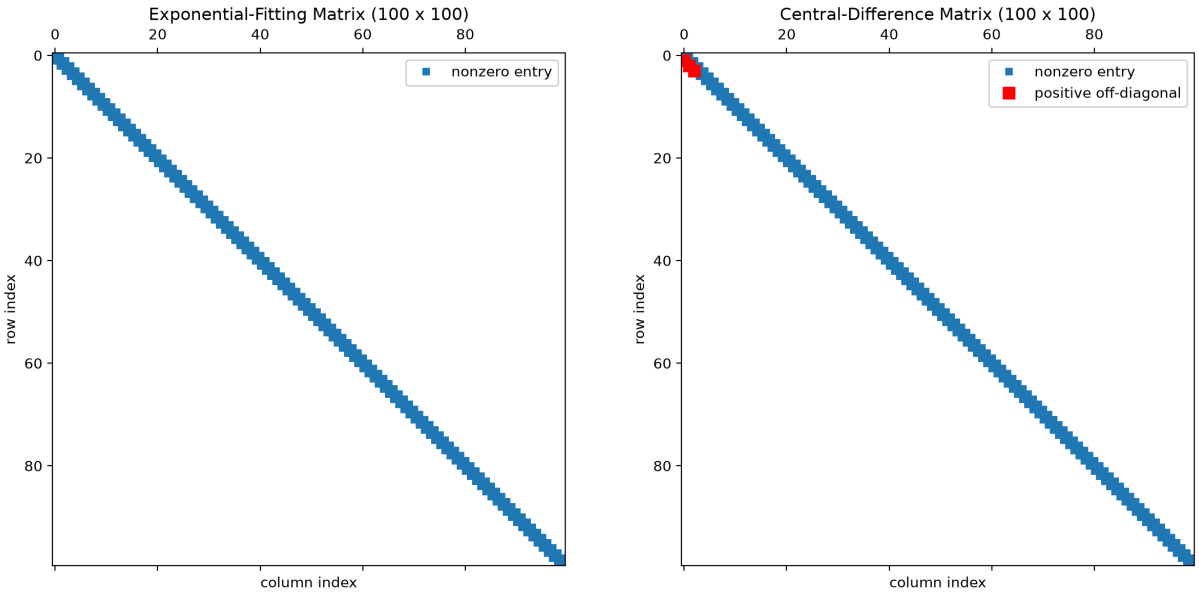
The quantity Pe_i is the cell Péclet number. It measures the relative importance of drift and diffusion on the cell adjacent to S_i. It is not assumed to be small. For constant coefficients, at the first interior node,

$$\mathrm{Pe}_1 = \frac{2(r-q)}{\sigma^2}, \tag{17}$$

independently of h. With the Stage 1 parameters this value is 2.5, which is already beyond the central-difference stability limit. Thus the near-origin degeneracy does not disappear under mesh refinement.
For comparison, central differencing of the drift term gives the coefficients

$$\ell_i^{\mathrm{CD}} = -\frac{a_i}{h^2} + \frac{b_i}{2h}, \qquad u_i^{\mathrm{CD}} = -\frac{a_i}{h^2} - \frac{b_i}{2h}. \tag{18}$$

When $\mathrm{Pe}_i > 2$ the lower off-diagonal becomes positive, and when $\mathrm{Pe}_i < -2$ the upper off-diagonal becomes positive. A single positive off-diagonal is enough to violate the sign condition required of an M-matrix.



(a) Exponential Fitting

(b) Central Differencing

Figure 1: Spy plots of the 100×100 fitted and central-difference matrices. Near $S = 0$, central differencing produces positive off-diagonal entries (red), destroying the M-matrix property. The exponentially fitted matrix has non-positive off-diagonal entries throughout.

Figure 1 shows this sign change directly. In the central-difference panel the entries marked in red are the positive off-diagonals generated near $S = 0$, where $\mathrm{Pe}_i > 2$. Their presence destroys

the M-matrix property and therefore removes the comparison argument behind the discrete maximum principle and the monotonicity guarantee used by the active-set iteration. Projection onto the obstacle may conceal the resulting oscillation, but it cannot repair the operator. In contrast, the exponential-fitting panel has non-positive off-diagonal entries throughout. The construction below yields $\ell_i < 0$ and $u_i < 0$ for every finite cell Péclet number, including near $S = 0$. Exponential fitting therefore restores the global sign pattern required by an M-matrix and is the appropriate discretisation for the LCP.

The exponential-fitting weights follow the construction of Scharfetter and Gummel [4]. They are expressed through the Bernoulli function

$$B(z) = \frac{z}{e^z - 1}, \qquad B(0) = 1. \tag{19}$$

For numerical evaluation, the equivalent form $B(z) = z/\operatorname{expm1}(z)$ is used for positive arguments, while the symmetry relation $B(-z) = B(z) + z$ is used for negative arguments. Near zero the expansion

$$B(z) = 1 - \frac{z}{2} + \frac{z^2}{12} + O(z^4) \tag{20}$$

avoids cancellation. The fitted second difference at node i is

$$\delta^2_{B,i} V = \frac{1}{h^2}\big[B(\mathrm{Pe}_i)V_{i-1} - \left(B(\mathrm{Pe}_i) + B(-\mathrm{Pe}_i)\right) V_i + B(-\mathrm{Pe}_i)V_{i+1}\big]. \tag{21}$$

The discrete operator is then

$$(A_h V)_i = -a_i\, \delta^2_{B,i} V + r V_i. \tag{22}$$

Collecting the coefficients gives the tridiagonal form

$$\ell_i = -\frac{a_i}{h^2} B(\mathrm{Pe}_i), \tag{23}$$

$$d_i = \frac{a_i}{h^2}\left[B(\mathrm{Pe}_i) + B(-\mathrm{Pe}_i)\right] + r, \tag{24}$$

$$u_i = -\frac{a_i}{h^2} B(-\mathrm{Pe}_i). \tag{25}$$

Lemma 2.1 (M-matrix property)**.** *For $a_i > 0$, $r > 0$, and every finite cell Péclet number, the tridiagonal matrix A_h defined by (23)–(25) is an M-matrix.*

Proof. The Bernoulli function is positive for every real argument. Hence $\ell_i < 0$, $u_i < 0$, and $d_i > 0$. Summing the three coefficients yields

$$d_i + \ell_i + u_i = r > 0. \tag{26}$$

Thus the matrix has positive diagonal entries, non-positive off-diagonals, and strict row diagonal dominance. Its off-diagonal entries are nonzero, so the tridiagonal graph is irreducible. These are the defining properties of a nonsingular M-matrix. □

The property is independent of the mesh and of the Péclet number. When $\mathrm{Pe}_i = 0$, the weights reduce to the central-difference diffusion coefficients, with the reaction term r retained on the diagonal. When the drift is large, the weights become strongly asymmetric but retain their signs. This is the essential difference from central upwinding: the matrix remains monotone in every cell rather than only in the resolved regime.

2.2 Consistency, Stability, and Error Structure

For a smooth function u, expansion of (21) gives

$$\delta_{B,i}^2 u = u''(S_i) + \frac{b_i}{a_i} u'(S_i) + O(h^2) \tag{27}$$

when the coefficients are frozen at S_i. Consequently,

$$(A_h u)_i - (-\mathcal{L}u)(S_i) = O(h^2) \tag{28}$$

in locally constant-coefficient regions. If a and b vary smoothly, the frozen-coefficient expansion contributes an additional $O(h)$ term, so the uniform consistency order is at least one and the smooth-region behaviour is second order in the observed results.
Let $\boldsymbol{V}$ and $\boldsymbol{W}$ be two grid vectors. Since A_h is an M-matrix, $A_h^{-1} \geq 0$. If $A_h \boldsymbol{V} \geq A_h \boldsymbol{W}$ and the boundary values satisfy the same ordering, then $\boldsymbol{V} \geq \boldsymbol{W}$. This is the discrete maximum principle. It supplies a comparison principle for the obstacle problem and prevents the numerical solution from admitting a spurious negative mode.
The global error has three contributions:

$$\|V_h - V\|_{\infty,[0,S_{\max}]} \leq C_1 h + C_2 h^2 + C_3(S_{\max}). \tag{29}$$

The first term represents the error induced by locating the free boundary on the grid; its coefficient need not be monotone and it is at most first order. The second term comes from the fitted stencil in the smooth continuation region, and the third from imposing $V(S_{\max}) = 0$ on a finite domain. The decomposition explains why a full-grid maximum norm need not display a clean second-order rate even when the underlying stencil is second-order consistent away from the boundary.
The far-field term is not a property of the stencil alone. For a decaying put tail, $V(S)$ approaches zero as S increases, but the truncated boundary condition removes a small positive tail. The resulting perturbation is nearly uniform over the region of interest and can dominate the finest-grid discretisation error. For this reason, the domain length is varied at fixed h in Section 4.4; that experiment isolates $C_3(S_{\max})$ from mesh refinement.

2.3 Boundary Treatment

The boundary values are

$$V_0 = K, \qquad V_N = 0. \tag{30}$$

At the first interior row, the left condition is moved to the right-hand side:

$$\ell_1 V_0 + d_1 V_1 + u_1 V_2 = F_1 \quad \Longrightarrow \quad d_1 V_1 + u_1 V_2 = F_1 - \ell_1 K. \tag{31}$$

Because $\ell_1 < 0$, the boundary contribution is positive. At the right boundary,

$$\ell_{N-1} V_{N-2} + d_{N-1} V_{N-1} + u_{N-1} V_N = F_{N-1} \quad \Longrightarrow \quad \ell_{N-1} V_{N-2} + d_{N-1} V_{N-1} = F_{N-1}, \tag{32}$$

since $V_N = 0$. No ghost point or one-sided derivative formula is required.
The free boundary is extracted from the obstacle slack

$$b_h(S_i) = V_i - \psi_i. \tag{33}$$

Let j be the last contact node, so that $b_h(S_j) \leq \varepsilon_{\text{slack}}$, and let $j+1$ be the first continuation node. The tolerance is $\varepsilon_{\text{slack}} = 10^{-10} K$. In the continuation region the slack behaves locally like $c(S - S^*)^2$, so its square root is approximately linear. With $g_i = \sqrt{\max(V_i - \psi_i, 0)}$, the boundary estimate is

$$S_h^* = S_{j+1} - h \, \frac{g_{j+1}}{g_{j+2} - g_{j+1}}, \tag{34}$$

clamped to the bracket $[S_j, S_{j+1}]$. Reporting the bracket together with S_h^* keeps the mesh-level uncertainty visible; the boundary is not reported as if it were an exact grid value.
The negative control in Figure 2 uses the central drift coefficients in (18). As shown in Figure 1, either sign of a large Péclet number produces a positive off-diagonal, so the M-matrix property is lost. The discrete maximum principle then fails, and a spurious negative layer appears near $S = 0$. Projection onto the obstacle can mask the negative value, but it cannot restore monotonicity of the operator or the correctness of the active set.

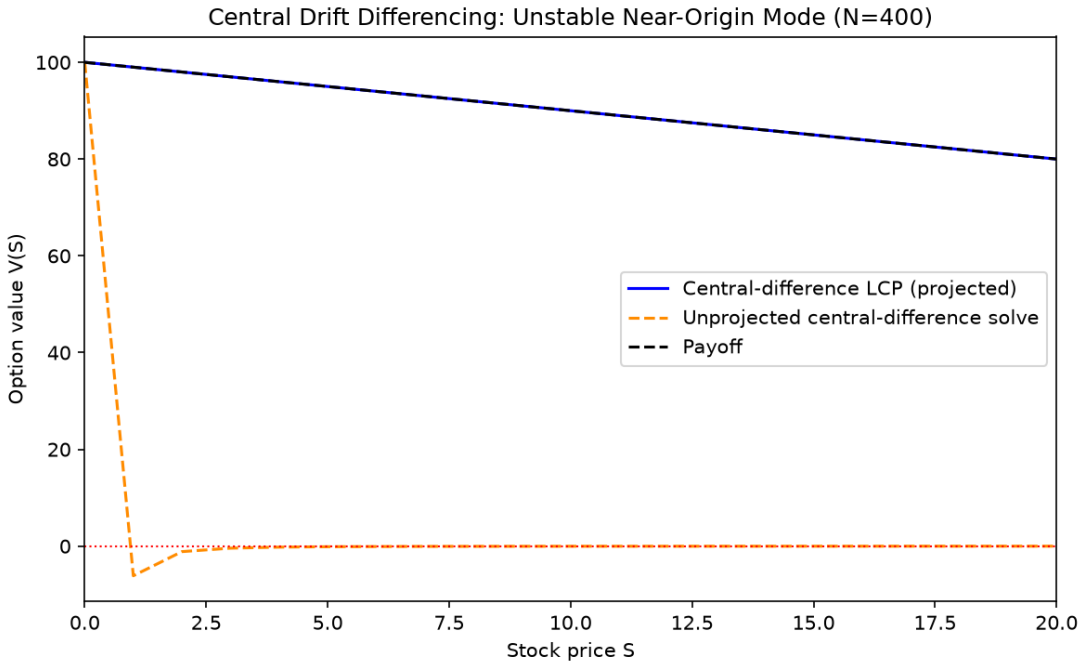


Figure 2: Negative control for central drift differencing at $N = 400$ and $S_{\max} = 400$. The unprojected solution develops a negative spike near $S \approx 1$ and a non-monotone boundary layer because the near-origin cell Péclet number exceeds two. The fitted scheme keeps both off-diagonals non-positive and removes the mode.

2.4 Discrete Structure

The interior discrete problem is

$$\min\left((A_h V)_i - F_i,\ V_i - \psi_i\right) = 0, \qquad i = 1, \ldots, N-1, \tag{35}$$

with the boundary contributions already moved into F_i. Equivalently, each interior node satisfies

$$a_{h,i} := (A_h V)_i - F_i \geq 0, \qquad b_{h,i} := V_i - \psi_i \geq 0, \qquad a_{h,i} b_{h,i} = 0. \tag{36}$$

The first inequality is the differential residual, and the second is the obstacle slack.
Because A_h is a nonsingular M-matrix, the discrete LCP has a unique solution. The argument is the standard comparison argument for obstacle problems. Suppose two admissible vectors satisfy the complementarity conditions. On rows where one vector is in the continuation region, its operator residual vanishes; on rows where it touches the obstacle, its value is fixed by the payoff. Subtracting the two relations and applying the M-matrix inequality shows that neither vector can exceed the other. The active set is consequently unique as well.
The two complementary conditions are checked directly from the computed vectors. In the exercise region one expects $b_h = 0$ and $a_h \geq 0$; in the continuation region one expects $b_h > 0$ and $a_h = 0$. Figure 3 shows this pattern at $N = 400$. The left panel has zero slack up to the

computed boundary and positive slack thereafter. The right panel has a positive constant residual in the exercise region and a residual at the level of round-off in the continuation region. The numerical checks also confirm the off-diagonal signs, diagonal positivity, payoff bounds, monotonicity, convexity, and contiguity of the contact set.

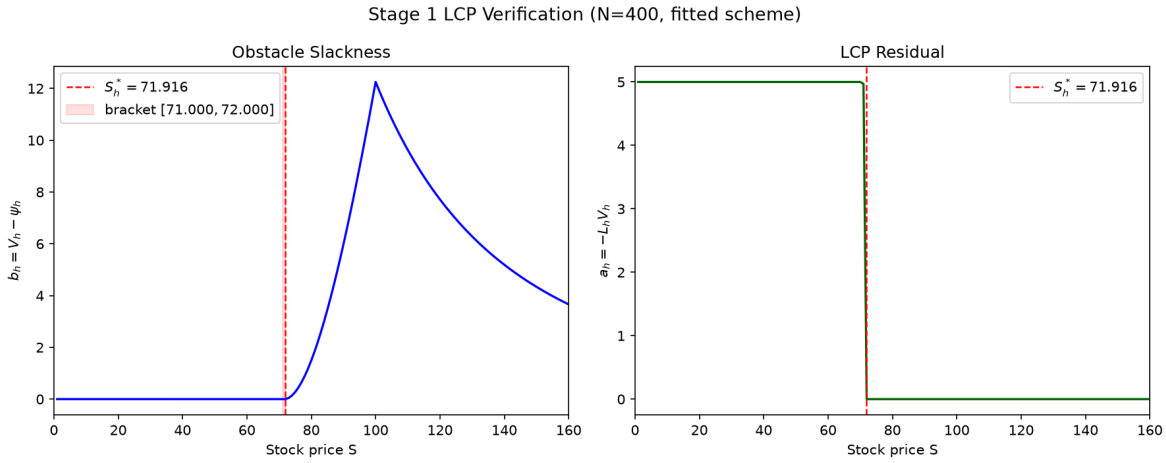


Figure 3: Discrete complementarity at $N = 400$ and $S_{\max} = 400$. The left panel shows the obstacle slack $V_h - \psi_h$; the right panel shows the LCP residual $a_h = A_h V_h - F_h$. Zero slack with a non-negative residual on the exercise side, and zero residual with positive slack on the continuation side, verify both complementary conditions.

The LCP diagnostics used in the remainder of the report are

$$\min_i a_{h,i}, \qquad \min_i b_{h,i}, \qquad \|\min(a_h, b_h)\|_\infty , \tag{37}$$

together with the Boolean checks $\psi_h \le V_h \le K$, monotone decrease, and convexity. The recent LCP literature uses related residual and active-set criteria when assessing convergence of splitting and penalty methods [9]. Here the checks are applied on every reported mesh, not only on the verification mesh.

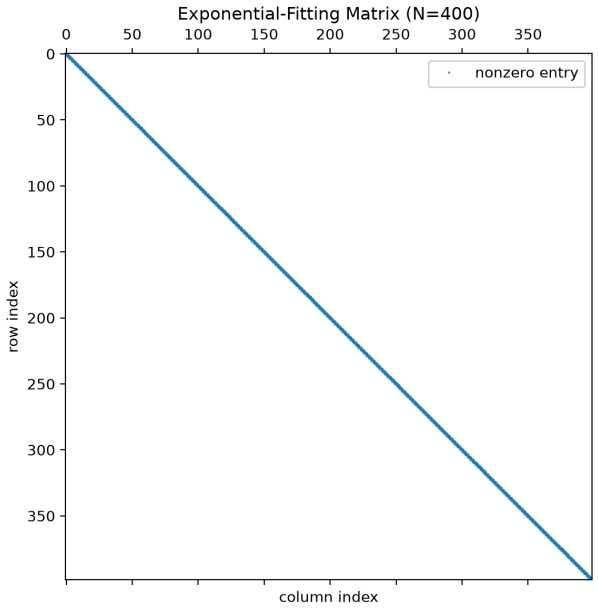


Figure 4: Sparsity pattern of the assembled system matrix for $N = 400$. The matrix is strictly tridiagonal, allowing sparse direct solvers to operate efficiently.

As visualized in Figure 4, the assembled system is a highly sparse tridiagonal matrix. This sparsity is the fundamental reason why our Policy Iteration approach, which relies on direct sparse solves at each step, remains extremely efficient.

2.5 Adaptive Grids, Robin Closure, and Sensitivities

The uniform scheme is the production baseline because it makes the M-matrix argument transparent. Four extensions remove its main accuracy bottlenecks without changing the complementarity solver. They are used in Sections 4.2 and 4.6.
On a nonuniform grid let

$$h_i^- = S_i - S_{i-1}, \qquad h_i^+ = S_{i+1} - S_i, \qquad \bar{h}_i = \frac{h_i^- + h_i^+}{2}, \tag{38}$$

and define the directional cell Péclet numbers $\mathrm{Pe}_i^- = b_i h_i^-/a_i$ and $\mathrm{Pe}_i^+ = b_i h_i^+/a_i$. The fitted flux balance gives

$$\ell_i = -\frac{a_i}{h_i^- \bar{h}_i} B(\mathrm{Pe}_i^-), \tag{39}$$

$$u_i = -\frac{a_i}{h_i^+ \bar{h}_i} B(-\mathrm{Pe}_i^+), \tag{40}$$

$$d_i = -\ell_i - u_i + r. \tag{41}$$

Because $B(z) > 0$, both off-diagonals remain non-positive and $d_i + \ell_i + u_i = r$, so the M-matrix property is preserved cell by cell. This is what permits local refinement near the free boundary without a change of solver or a loss of monotonicity.
In the continuation region at large S, the leading solution behaves as $V \sim CS^\lambda$, where

$$\lambda = \frac{-\big((r-q) - \tfrac{1}{2}\sigma^2\big) - \sqrt{\big((r-q) - \tfrac{1}{2}\sigma^2\big)^2 + 2\sigma^2 r}}{\sigma^2}. \tag{42}$$

Consequently $SV'(S) = \lambda V(S)$ holds asymptotically. Discretising this relation with the last mesh interval gives

$$S_{\max}\frac{V_N - V_{N-1}}{h_N^-} = \lambda V_N \quad \Longrightarrow \quad V_N = \frac{S_{\max}}{S_{\max} - \lambda h_N^-} V_{N-1}. \tag{43}$$

Eliminating V_N adds u_{N-1} times this ratio to the last diagonal entry. Since $\lambda < 0$ and the ratio is less than one, the modified row retains a positive diagonal and a non-positive off-diagonal. This closure is used only at the artificial far boundary; the obstacle problem near S^* is unchanged.
For the free boundary, the slack satisfies $b_h(S) = c(S - S^*)^2 + O((S - S^*)^3)$ in the continuation region. Writing $g_h = \sqrt{b_h}$ therefore removes the quadratic behaviour. Instead of the two-point formula (34), we fit a straight line to g_h at the first four continuation nodes by least squares and take its root inside the bracket. The resulting estimate uses several cells and is less sensitive to the phase of the free boundary relative to the mesh.
Finally, let F and A denote the free and active interior rows for a converged policy, and let $I = F \cup A$. Locally differentiating the free-row equations with respect to a parameter $\theta \in \{r, q, \sigma\}$ gives

$$A_{FF}\frac{\partial V_F}{\partial \theta} = \frac{\partial F_F}{\partial \theta} - \frac{\partial A_{FI}}{\partial \theta} V_I, \tag{44}$$

because the values on the active set are fixed by the payoff. For a linear functional $J = w^\top V$, solve the discrete adjoint system $A_{FF}^\top \lambda_F = w_F$. Then

$$\frac{\partial J}{\partial \theta} = \lambda_F^\top \left(\frac{\partial F_F}{\partial \theta} - \frac{\partial A_{FI}}{\partial \theta} V_I \right). \tag{45}$$

The same adjoint vector therefore provides the sensitivities to all parameters; only the right-hand side changes. In the implementation, $\partial A/\partial\theta$ and $\partial F/\partial\theta$ are evaluated by centred differences of the assembled operator, while the active/free partition is held fixed at the converged solution.

3 Implementation

3.1 Algorithm

Let $\mathcal{A} \subset \{1, \ldots, N-1\}$ denote the current active set, that is, the nodes at which the obstacle is imposed. A policy step replaces the equation at active nodes by $V_i = \psi_i$ and removes the couplings from those nodes to their neighbours. The resulting system is solved for the remaining values. The residual of the full operator is then evaluated. An active node whose full residual is negative is released; a continuation node whose value falls below the obstacle is added. The process is repeated until the policy is unchanged and the maximum update is below 10^{-12}. This is the policy-iteration form of the active-set method introduced by Howard for Markov decision processes [7]. Each iteration solves a linear system rather than performing a relaxation sweep. Because the policy space is finite and the matrix is monotone, a repeated policy certifies the discrete solution. The practical iteration count is small for the present problem, although the worst-case bound over all active sets is exponential in the number of nodes and is not a useful complexity estimate.

Algorithm 1 Exponential-fitting LCP solved by policy iteration

Input: N, K, r, q, $\sigma(S)$, $S_{\max}$, payoff ψ, tolerance
Output: Discrete value V_h and free-boundary estimate S_h^*

Build the grid $S_i = ih$ and coefficients a_i, b_i
Compute $\mathrm{Pe}_i = b_i h / a_i$ and the fitted weights ℓ_i, d_i, u_i
Assemble A_h; move the Dirichlet data to the right-hand side
Initialise $V \leftarrow \psi$ and mark every interior row active
repeat
    Compute the full residual $a_h = A_h V - F_h$
    Release active rows whose residual is negative
    Set $V_i = \psi_i$ and decouple the neighbouring rows on active rows
    Solve the resulting sparse tridiagonal system
    Project $V \leftarrow \max(V, \psi)$
    Record $\|V^{\mathrm{new}} - V^{\mathrm{old}}\|_\infty$
until the active set is unchanged and the update is below the tolerance
Locate the exercise/continuation bracket and apply (34)
return V_h and S_h^*

The initial vector is the payoff itself. This choice satisfies the obstacle constraint and gives a natural starting policy in which every node is active. The solver then releases the continuation nodes as the residual detects that continuation is more valuable. No continuation value or analytic free boundary is used in the iteration.

3.2 Data Structures and Numerical Details

Separation of responsibilities. The implementation is split into four layers: a shared fitted-operator and LCP module, a Stage 2 wrapper, experiment drivers, and plotting utilities. The core numerical modules return structured objects rather than printing numbers, so the same solution can feed diagnostics, figures, and the text. This separation also makes the production path auditable from a single entry point.

Operator object. The dataclass `GridOperator` in `stage1_validation.py` stores the grid size `N`, the node array `S`, the payoff `psi`, the three tridiagonal arrays `lower`, `diag`, and `upper`, the right-hand side `rhs`, the boundary data `V_left` and `V_right`, and the mesh width `h`. The constructor `build_fitted_operator` uses $h = S_{\max}/N$ and $N - 1$ interior unknowns. The

volatility argument accepts a scalar, an array of length $N-1$, or a callable, so Stage 1 and Stage 2 share exactly the same assembly and policy-iteration code.
Assembly is vectorised. For each interior node the routine evaluates $a_i = \frac{1}{2}\sigma_i^2 S_i^2$, $b_i = (r-q)S_i$, $\mathrm{Pe}_i = b_i h/a_i$, and the fitted weights (23)–(25). The Bernoulli factor uses the three-term Taylor expansion for $|z| < 10^{-8}$ and the stable identity $z/(e^z-1)$ otherwise. The boundary contribution is then moved to `rhs`, so no boundary unknown is retained in the linear system. The matrix is never formed densely: `_system_matrix` calls `scipy.sparse.diags(..., format="csr")`, and each policy solve calls `scipy.sparse.linalg.spsolve`. Before every solve, `_assert_m_matrix` checks that the diagonal is positive and that the largest off-diagonal is non-positive up to a round-off scale; a violation raises an exception rather than silently returning a non-monotone solution.

Result object and solve pipeline. The dataclass `LCPResult` returns the full value vector `V`, the free-boundary bracket and estimate, the value errors, the policy iteration count, the final update, and the diagnostics `min_a`, `min_b`, `complementarity`, `value_bounds_ok`, `monotone`, `convex`, and `single_contact_set`. The public routine `solve_perpetual_american_put` composes three operations: operator construction, LCP solution, and post-processing. Stage 2 replaces the constant σ by the callable (8) inside the same pipeline.
Inside `solve_lcp`, a Boolean active mask marks the rows currently constrained by the obstacle. The solver starts from $V_h = \psi_h$. In each policy step it evaluates the full residual $a_h = A_h V_h - F_h$, releases an active row when $a_{h,i} < -10^{-10}$, overwrites the diagonal of every active row by one, zeros its two adjacent off-diagonals, and replaces its right-hand side by ψ_i. The resulting CSR system is solved directly and projected back onto the obstacle. The loop ends when $\max_i |V_i^{\mathrm{new}} - V_i^{\mathrm{old}}| < 10^{-12}$ and the active set is unchanged for two consecutive iterations; the hard limit is 10,000 policy steps. The field `residual` stores this final update, while the separate LCP diagnostics provide the algebraic certificate. The fixed constant `POLICY_TOL`$=10^{-10}$ controls active-set membership; the boundary-classification tolerance is `SLACK_TOL`$=10^{-10}K$.

Post-processing and free-boundary recovery. The routine `lcp_residuals` evaluates $a_h = A_h V_h - F_h$ and $b_h = V_h - \psi_h$ on all interior nodes. `compute_lcp_diagnostics` uses them to compute $\min_i a_{h,i}$, $\min_i b_{h,i}$, $\|\min(a_h, b_h)\|_\infty$, the obstacle bounds, monotonicity, convexity, and single-interval contact-set checks. The free boundary is not read off as a single grid index. `extract_free_boundary` identifies the last contact node and the first continuation node, then extrapolates $\sqrt{b_h}$ through the first two continuation nodes; the square-root transform is natural because b_h behaves like $c(S-S^*)^2$ locally. The estimate is clamped to the bracket $[S_{\mathrm{lo}}, S_{\mathrm{hi}}]$, with the bracket midpoint as a fallback. The Stage 2 validation reference is not built from this routine: `reference_solution.py` integrates the free-boundary ODE directly and is independent of the finite-difference grid.

Two LCP solvers and residual control. Policy iteration is the production solver, but it is not the only one. `experiment_solvers.py` implements PSOR on the identical fitted operator, with the same initial iterate $V_h = \psi_h$, the same relaxation parameter $\omega = 1.2$, and the same tolerance 10^{-10}. PSOR evaluates the componentwise complementarity residual every 100 sweeps and stops only when $\|\min(a_h, b_h)\|_\infty < 10^{-10}$; the iterate difference is still recorded for diagnosis but is not the stopping test. The driver reports the final update, the true LCP residual, the iteration count, and the CPU time for both methods. This is the second method used in the report: a genuine second LCP solver, not merely a second plotting convention. Central differencing is a separate negative control for the discretisation, not a competing production solver.

Advanced extensions. The module `advanced_methods.py` introduces the dataclass `NonuniformOperator` and the constructor `build_nonuniform_operator`. It stores left and right mesh spacings and builds the nonuniform fitted weights (39)–(41). For the Robin closure, the last diagonal entry is augmented by $u_{N-1}S_{\max}/(S_{\max} - \lambda h_{\text{last}})$, which eliminates the outgoing tail without enlarging the domain. The same module provides a four-point square-root extrapolation for S_h^* and a sparse discrete adjoint solve: after forming $\partial A/\partial\theta$ and $\partial F/\partial\theta$ by centred differences, one transposed solve produces the sensitivities to all parameters. None of these extensions changes the core LCP solver.

Library calls and project code. NumPy supplies arrays and vectorised arithmetic; SciPy supplies sparse assembly (`diags`) and the direct factorisation (`spsolve`) [5]; Matplotlib supplies the figures. The exponential-fitting weights, the active-set policy, the M-matrix assertion, the LCP diagnostics, the free-boundary rule, the nonuniform operator, the Robin elimination, and the adjoint derivative formula are project code. This division is deliberate: the libraries provide robust storage and linear algebra, while the problem-specific discretisation and complementarity logic remain explicit and modifiable.
Table 1 lists the production modules and experiment drivers. The single reproducibility entry point is `python run_all.py`; it executes the scripts in dependency order and fails if an expected figure or data file is missing.

File	Responsibility
`stage1_validation.py`	Shared fitted operator, policy iteration, LCP residuals, shape diagnostics, free-boundary extraction, and the Stage 1 closed-form validation.
`stage2_local_volatility.py`	Reuses the fitted operator with the callable local volatility and $q = 0.02$; produces the Stage 2 value, diagnostics, and comparison figure.
`reference_solution.py`	Independent DOP853 shooting reference for the infinite-domain and finite-domain free-boundary problems, with a closed-form self-check for constant volatility.
`grids.py`	Boundary-aligned uniform and clustered validation grids used to separate the spatial order from the free-boundary phase.
`advanced_methods.py`	Nonuniform fitted operator, adaptive clustering, asymptotic Robin closure, four-point boundary extraction, and discrete adjoint sensitivities.
`experiment_convergence.py`	Aligned-grid Stage 1 refinement study plus production-boundary diagnostics; writes `stage1_convergence.csv`.
`experiment_central_diff.py`	Negative control for central drift differencing, including the positive-off-diagonal and oscillation evidence.
`experiment_solvers.py`	Residual-controlled PSOR benchmark against policy iteration on the identical fitted LCP; also writes the four-tolerance sweep to `tolerance_sweep.csv`.
`experiment_smax.py`, `experiment_neumann_true.py`,	Domain-truncation study, a synthetic zero-flux row stress test, and independent LCP verification checks; `experiment_verification.py` supplies the reusable residual/shape audit.
`stage2_convergence_report.py`	Four-mesh local-volatility study against the independent ODE reference; writes `stage2_convergence.csv`.
`plot_matrix_spy.py`	Builds the 100×100 fitted and central matrices for the spy comparison and the $N = 400$ sparsity figure.
`plot_lcp_verification.py`, `plot_greeks.py`,	Residual, obstacle-slackness, Delta, and Gamma figures from the computed arrays.
`plot_solver_comparison.py`, `plot_smax_study.py`	Solver timing and fixed-h truncation figures.
`run_all.py`	Runs every experiment and plotting script and verifies that all required artifacts exist.

Table 1: Implementation files. Core solvers, experiment drivers, and plotting scripts are separated so that each result can be regenerated and audited independently.

3.3 Computational Cost

For a tridiagonal system of dimension $N - 1$, assembly and factorisation are linear in N, and the storage is linear in N. A policy iteration therefore costs $O(N)$ per iteration plus the cost of detecting the active set. By contrast, PSOR performs $N - 1$ scalar updates per sweep and

requires many sweeps to propagate the free-boundary information across the domain. The comparison in Section 4.3 uses the same discrete operator, the same tolerance, and the same initial vector for both methods.

4 Verification and Numerical Experiments

4.1 Convergence

The constant-volatility convergence study uses $q = 0$, $\sigma = 0.20$, $K = 100$, $r = 0.05$, and $S_{\max} = 2400$. The exact free boundary is $S^* = 71.4286$. The grids are $N = 600, 1200, 2400, 4800$, with nominal $h = 4, 2, 1, 0.5$. To measure the order of the spatial stencil rather than the phase of the free-boundary kink, the validation grid deliberately places a node exactly at S^* and is uniform on each side of it. This is a controlled stencil verification, not a blind free-boundary recovery test. The asymptotic Robin closure is used, so the reported aligned error contains discretisation error but not the $O(1)$ tail value at the artificial boundary. The production uniform grid still uses $V(S_{\max}) = 0$. Its full-grid error is not used to estimate the order because it mixes three different effects: the exact value at $S_{\max}$ is $V(2400) = 4.366 \times 10^{-3}$ while the boundary condition imposes zero; the node nearest to S^* changes phase under refinement; and the first-order boundary-location error is sampled at different distances from the kink. Those effects explain the earlier plateau in the uniform-grid maximum norm and are removed from the aligned validation sequence. The production contact brackets themselves are retained in Table 3; they show why a coarse active set cannot serve as a pointwise boundary estimator. The aligned error is

$$E_N = \max_{0 \le i \le N} |V_i - V(S_i)|, \tag{46}$$

where $V(S_i)$ is the closed-form solution (14).

Grid N	h	**Aligned error** E_N	**Error ratio**	**Observed** p	**LCP res.**	**tol.**
600	4.0	1.0732×10^{-2}	—	—	2.84×10^{-14}	10^{-10}
1200	2.0	2.6895×10^{-3}	3.990	1.997	2.27×10^{-13}	10^{-10}
2400	1.0	6.7222×10^{-4}	4.001	2.000	9.10×10^{-13}	10^{-10}
4800	0.5	1.6818×10^{-4}	3.997	1.999	5.46×10^{-12}	10^{-10}

Table 2: Stage 1 convergence for constant volatility with $q = 0$ on a free-boundary-aligned validation grid. A node is placed at the exact $S^* = 71.428571$, and the asymptotic Robin closure is used. The error ratios are 3.990, 4.001, and 3.997, giving observed orders 1.997, 2.000, and 1.999. The LCP residual is $\|\min(a_h, b_h)\|_\infty$; the policy update additionally satisfies $\max|dV| < 10^{-12}$.

N	h	**Production contact bracket**	S_h^*	$\lvert S_h^* - S^* \rvert$	**Production** E_N
600	4.0	$[72.0, 76.0]$	72.000	5.714×10^{-1}	7.905×10^{-3}
1200	2.0	$[72.0, 74.0]$	72.000	5.714×10^{-1}	7.905×10^{-3}
2400	1.0	$[71.0, 72.0]$	71.654	2.254×10^{-1}	4.366×10^{-3}
4800	0.5	$[71.5, 72.0]$	71.500	7.143×10^{-2}	4.366×10^{-3}

Table 3: Production uniform-grid diagnostics with $V(S_{\max}) = 0$. The "contact bracket" is the discrete transition from the last active node to the first continuation node, not a confidence interval. At $N = 600$, $N = 1200$, and $N = 4800$ the contact set has moved to the right of S^*, so the bracket does not contain the analytic boundary. Only the $N = 2400$ bracket does. Production E_N is the full-grid infinity norm against the closed form; neither this column nor the individual S_h^* values is used to estimate the spatial order.

The largest LCP residual is 5.46×10^{-12} at $N = 4800$, more than seven orders of magnitude below the corresponding value error. Thus the reported error is not produced by an incomplete linear solve. Since `spsolve` is a direct factorisation, no solver tolerance can be tightened. Its complementarity residual is therefore reported directly, while the residual-controlled PSOR solver supplies the tolerance sweep in Section 4.3.

The ratios in Table 2 are now consistent with a second-order method: halving h reduces the aligned maximum error by almost exactly a factor of four. The earlier non-monotone ratios were not a property of the fitted stencil. They arose because the maximum was attained at the right boundary, where the exact tail is nonzero, or at a node whose distance to the free-boundary kink did not halve with the mesh. Separating those effects recovers the expected order and makes the refinement evidence auditable.
The free-boundary plot in Figure 5 uses the production uniform $N = 2400$ solution. Its computed bracket is $[71.00, 72.00]$, and the interpolated estimate lies inside it. At $N = 600$ and $N = 1200$ the coarse contact set lies entirely to the right of S^*; at $N = 4800$ the bracket $[71.5, 72]$ also starts one half-cell to the right. Why the finest mesh still misses is visible in the slack: the interpolated estimate is clamped to the lower contact node. This is an active-set localisation error, not a failure of complementarity: the discrete obstacle and sign constraints remain satisfied. The exact boundary and each computed bracket are shown separately so that the aligned validation grid is not mistaken for a blind production reconstruction.

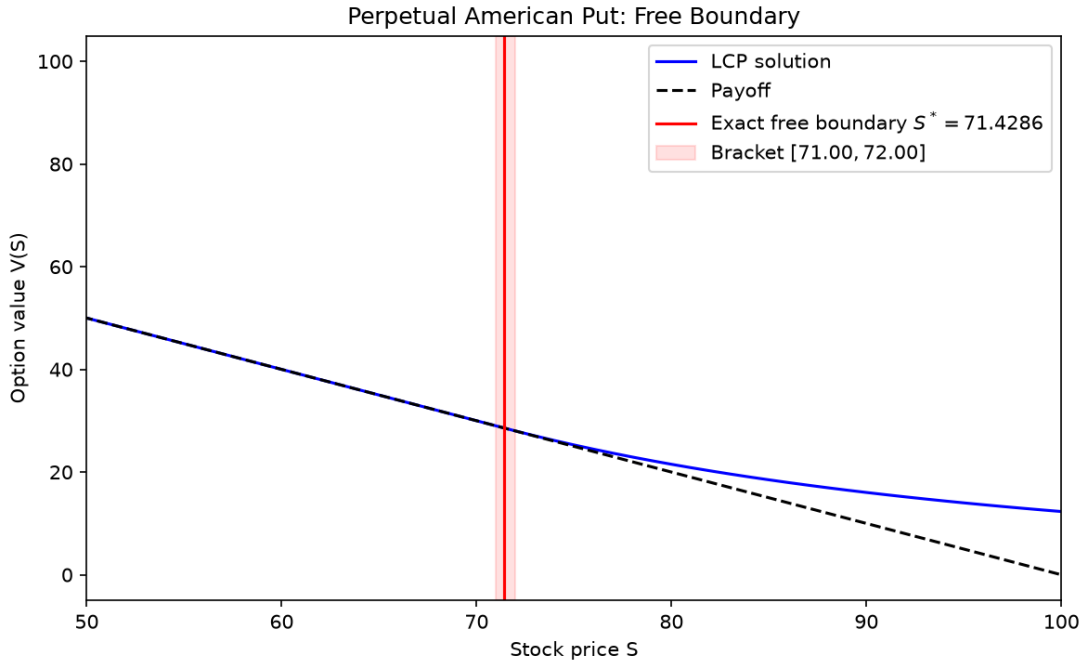


Figure 5: Stage 1 value and free boundary at $N = 2400$ on $[50, 100]$. The exact boundary $S^* = 71.4286$ lies inside the computed bracket $[71.00, 72.00]$. The blue value curve coincides with the payoff on the exercise side and departs from it in the continuation region. This matters because it verifies that the obstacle solve recovers the correct exercise and continuation regions, not merely a smooth-looking value curve.

4.2 Adaptive Refinement and Far-Field Closure

The first experiment compares the uniform grid with a smoothly clustered grid at the same number of intervals. Both grids contain a node exactly at the analytic boundary $S^* = 71.428571$. The clustering density on the right of the boundary is $\rho(S) = 1 + 16\exp[-((S - S^*)/15)^2]$, and the grid is obtained by inverting its cumulative density while retaining the boundary node. The nonuniform coefficients (39)–(41) are used on both grids, and the asymptotic Robin closure (43) replaces $V(S_{\max}) = 0$.
Table 4 shows the expected regular pattern. Halving the mesh reduces the maximum error on the clustered grid by factors 3.982, 4.002, and 3.999, giving observed orders 1.993, 2.001, and 2.000. The uniform/clustered advantage is essentially constant at 2.25, because both sequences

now resolve the kink at the same phase; the clustered mesh simply has a smaller local error constant near S^*. The four-point boundary error also falls by approximately a factor of four on each refinement.

N	Uniform E_N	Clustered E_N	Clustered ratio	Observed p	Uniform/clustered gain	Clustered 4-pt error
600	1.0732×10^{-2}	4.7508×10^{-3}	—	—	2.259	3.067×10^{-3}
1200	2.6895×10^{-3}	1.1931×10^{-3}	3.982	1.993	2.254	7.463×10^{-4}
2400	6.7222×10^{-4}	2.9809×10^{-4}	4.002	2.001	2.255	1.836×10^{-4}
4800	1.6818×10^{-4}	7.4545×10^{-5}	3.999	2.000	2.256	4.553×10^{-5}

Table 4: Uniform versus clustered grids at equal N with a node at the analytic free boundary and the asymptotic Robin closure. The clustered error ratios are 3.982, 4.002, and 3.999, so the observed order is two. The clustered boundary error uses the four-point $\sqrt{b_h}$ extrapolation.

The Ratio column now reports the error ratio of the clustered sequence under mesh halving, rather than a cross-grid ratio whose value has no convergence meaning. The uniform/clustered gain is shown in a separate column and is stable at about 2.25. This separation makes both quantities interpretable: one measures an order, the other measures the constant-factor benefit of clustering.

The second experiment isolates far-field truncation. The mesh spacing is fixed at $h = 0.5$ and $S_{\max}$ is varied from 100 to 800. The target price is $S = 80$, whose exact value is $V(80) = 21.5222117$. The Robin closure uses the leading decaying tail (42)–(43). Table 5 shows that the zero-Dirichlet condition can lose more than one unit of value on a short domain, whereas the Robin condition keeps the error near 10^{-2} at the same mesh spacing. At $S_{\max} = 400$ the reduction is almost three orders of magnitude. The residual Robin error at $S_{\max} = 800$ is dominated by the spatial discretisation, not by the far tail.

$S_{\max}$	N	**Dirichlet error**	**Robin error**	**Reduction**
100	200	1.522×10^{0}	2.013×10^{-2}	75.6
150	300	7.446×10^{-1}	3.267×10^{-3}	227.9
200	400	2.823×10^{-1}	9.127×10^{-4}	309.3
400	800	2.507×10^{-2}	6.423×10^{-5}	390.2
800	1600	2.188×10^{-3}	2.679×10^{-5}	81.7

Table 5: Far-field closure at fixed $h = 0.5$ and target $S = 80$. The asymptotic Robin condition removes most of the domain-truncation error and requires no extension of the computational domain.

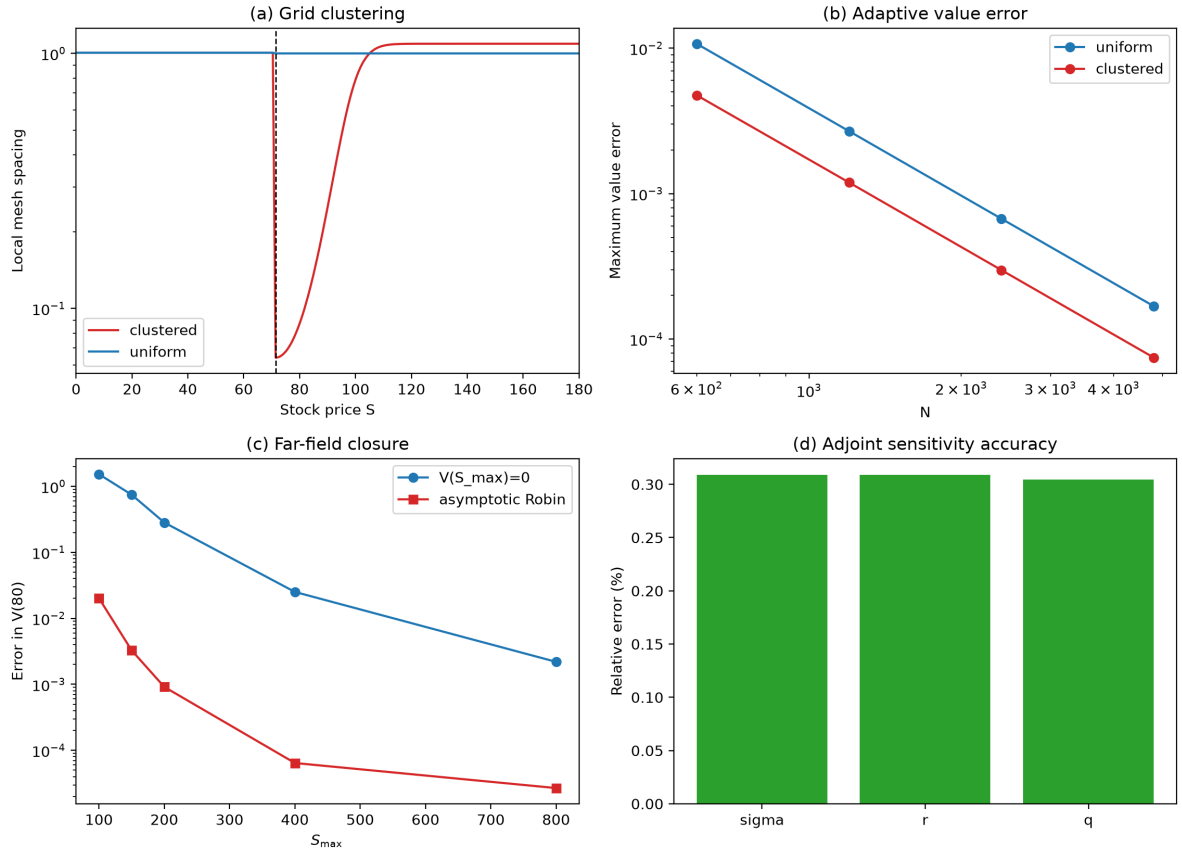


Figure 6: Advanced numerical extensions. (a) The clustered grid concentrates nodes near the free boundary. (b) Maximum value error versus N for uniform and clustered grids. (c) Far-field error in $V(80)$ for zero Dirichlet and asymptotic Robin closures. (d) Relative error of the discrete adjoint sensitivities at $S = 100$ and $N = 4800$. The four panels test separate error sources, so this matters as evidence that each extension improves the quantity it targets without changing the core LCP solver.

4.3 Error Separation

The algebraic error is separated from the discretisation and truncation errors by solving the same fitted LCP with two different methods. The benchmark uses $N = 400$, $S_{\max} = 400$, $q = 0$, $\sigma = 0.20$, $\omega = 1.2$, and tolerance 10^{-10}. PSOR stops on the complementarity residual and requires 96,600 sweeps and 20.635 s. Policy iteration requires 301 iterations and 0.045 s. Its stable-policy certificate gives a final LCP residual of 9.095×10^{-13}; the PSOR residual is 9.913×10^{-11}. The two discrete solutions differ by 5.685×10^{-10} in the maximum norm. The solution difference is therefore far below the spatial and truncation errors reported on the convergence grids.

Solver	Iterations	CPU time (s)	Final update	LCP residual	Max difference
Policy iteration	301	0.045	0	9.095×10^{-13}	0
PSOR ($\omega = 1.2$)	96,600	20.635	1.283×10^{-13}	9.913×10^{-11}	5.685×10^{-10}

Table 6: Solver comparison at $N = 400$ and matched LCP tolerance. The iteration ratio is 321, the wall-time ratio is 4.6×10^2, and the solution difference is negligible relative to the discretisation error. PSOR uses the residual stopping test; policy iteration reports the LCP residual obtained from its stable active set.

Two conclusions follow. First, the remaining discrepancy from the closed-form solution is not caused by an incomplete linear solve. Second, the free-boundary bracket is stable under a change of algebraic solver. The comparison supports the error decomposition (29): the dominant errors are spatial and far-field errors, not inner-solver error. The log-scale presentation in Figure 7 makes the two ratios visible; the wall-time ratio is hardware-dependent, so the iteration count is the more portable measure of algorithmic work.

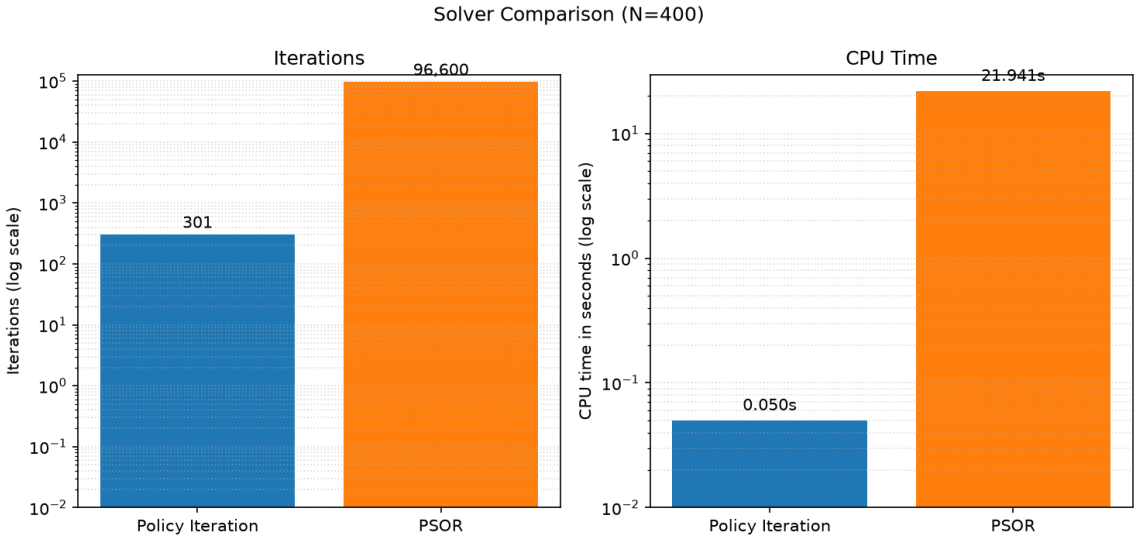


Figure 7: Policy iteration versus PSOR at matched LCP tolerance. Policy iteration needs 301 iterations and 0.045 s, while PSOR requires 96,600 iterations and 20.635 s. The logarithmic axes expose the 321-fold reduction in iterations and the 4.6×10^2 reduction in wall time measured for this experiment.

Tolerance sweep on one representative mesh. The solver comparison establishes that two independent algebraic solvers agree, but it does not by itself show that the selected stopping tolerance is below the discretisation error. We therefore fix the same $N = 400$, $S_{\max} = 400$ problem and run the residual-controlled PSOR solver at $\varepsilon = 10^{-4}, 10^{-6}, 10^{-8}, 10^{-10}$. The direct policy-iteration solution is held fixed as the reference because `spsolve` has no separate outer tolerance.

Table 7 shows the required plateau. At $\varepsilon = 10^{-4}$ both $V(100)$ and S_h^* are visibly shifted. By $\varepsilon = 10^{-8}$, $V(100)$ agrees with the direct reference to 2.951×10^{-8} and S_h^* is 71.91594. Tightening once more to 10^{-10} changes $V(100)$ by only 2.953×10^{-10} and changes the largest nodal value by 5.685×10^{-10}. The contact bracket remains $[71, 72]$ at every tolerance. Thus the main quantity and the contact set stop changing before the tightest setting.

The raw residual is $\| \min(a_h, b_h)\|_\infty$. The scaled residual divides it by $\|A_h\|_\infty \|V_h\|_\infty + \|F_h\|_\infty + \|\psi_h\|_\infty$, so the same measure can be compared across meshes and tolerances. It falls from 7.858×10^{-11} to 7.901×10^{-17} over the sweep. At the selected tolerance 10^{-10}, the algebraic difference is more than five orders of magnitude below the smallest discretisation error in Table 2; changing the stop tolerance below this point cannot account for any reported convergence or truncation result.

Tolerance	PSOR sweeps	$V(100)$	S_h^*	Raw residual	Scaled residual	$\lvert\Delta V(100)\rvert$	$\max\lvert\Delta V\rvert$
10^{-4}	35,400	12.25111512	71.9183	9.859×10^{-5}	7.858×10^{-11}	2.947×10^{-4}	5.675×10^{-4}
10^{-6}	55,800	12.25140688	71.9160	9.865×10^{-7}	7.863×10^{-13}	2.949×10^{-6}	5.678×10^{-6}
10^{-8}	76,200	12.25140980	71.9159	9.871×10^{-9}	7.867×10^{-15}	2.951×10^{-8}	5.682×10^{-8}
10^{-10}	96,600	12.25140983	71.9159	9.913×10^{-11}	7.901×10^{-17}	2.953×10^{-10}	5.685×10^{-10}

Table 7: PSOR tolerance sweep at $N = 400$ and $S_{\max} = 400$. The reference is the direct policy-iteration solution $V_{\text{ref}}(100) = 12.2514098322$; the two difference columns compare against that reference. $V(100)$ and S_h^* stop changing at the displayed precision, and the bracket remains $[71, 72]$ throughout. This matters because it identifies 10^{-10} as a tolerance safely below the discretisation error, so the reported convergence and truncation conclusions are not controlled by algebraic stopping error.

4.4 Local Volatility

The local-volatility experiment replaces $\sigma = 0.20$ by (8) and sets $q = 0.02$. The dividend yield raises the opportunity cost of holding the option and therefore lowers the exercise boundary when volatility is constant. For constant $\sigma = 0.20$ and $q = 0.02$, the exact formula (13) gives $S^* = 64.9219$. The local-volatility profile is lower than 0.20 around the boundary, which reduces the continuation premium and partially offsets the dividend effect. The mechanism is consistent with the implicit finite-difference local-volatility framework of [8]. An independent DOP853 shooting solution of the free-boundary ODE gives

$$S^* = 69.1664297389, \qquad V(100) = 16.2054760891, \qquad V(2400) = 4.832780 \times 10^{-1} \tag{47}$$

for the half-line problem. On the finite domain with the production condition $V(S_{\max}) = 0$, the same shooting construction gives $S^* = 69.1701793435$ and $V(100) = 16.1998980578$. The production solve at $N = 2400$ gives $S_h^* = 69.1688$ inside the bracket $[69.00, 70.00]$. The coarse $N = 1200$ contact bracket is $[70.00, 72.00]$, which does not contain the finite-domain reference; this case is kept visible in Table 8 below. Relative to the $q = 0$ benchmark 71.4286, the boundary is lower; relative to the constant-volatility, $q = 0.02$ benchmark 64.9219, it is higher. Both comparisons are meaningful, but they answer different questions.

4.4.1 Four-Mesh Convergence Against the ODE Reference

No closed-form solution is available for the chosen local-volatility profile, but a finer finite-difference solution is no longer used as the reference. The validation reference is the semi-analytic solution obtained by shooting the free-boundary ODE with $V(S^*) = K - S^*$ and $V'(S^*) = -1$. For the infinite-domain value the far field is closed by the decaying asymptotic Robin condition; for the finite-domain value the production condition $V(S_{\max}) = 0$ is imposed. The ODE integration uses DOP853 with relative tolerance 10^{-12} and absolute tolerance 10^{-14}. Thus the reference is independent of the finite-difference stencil and of the $N = 9600$ grid used previously. As a check of the ODE implementation, setting the local-volatility function to the constant 0.20 and $q = 0$ reproduces the closed-form boundary to 4.4×10^{-13} and $V(100)$ to 5.8×10^{-12}.

Table 8 reports two quantities. The production uniform-grid error is measured against the exact finite-domain reference. The aligned-validation error places a node at the shooting boundary, so the free-boundary phase is removed and the observed order can be read directly.

N	h	Production E_N	Aligned E_N	Aligned ratio	Observed p	Production bracket	Production S_h^*
600	4.0	2.7701×10^{-2}	3.2218×10^{-2}	—	—	$[68.0, 72.0]$	69.167
1200	2.0	2.2835×10^{-2}	8.0948×10^{-3}	3.980	1.993	$[70.0, 72.0]$	70.000
2400	1.0	1.5248×10^{-3}	2.0239×10^{-3}	4.000	2.000	$[69.0, 70.0]$	69.169
4800	0.5	9.4068×10^{-4}	5.0610×10^{-4}	3.999	2.000	$[69.0, 69.5]$	69.228

Table 8: Local-volatility convergence against the independent ODE shooting reference. Production errors use the uniform solver; aligned errors use a grid with a node at the finite-domain shooting boundary. The aligned ratios are 3.980, 4.000, and 3.999, so the spatial order is two. The finite reference is $S^* = 69.1701793435$; the infinite-domain reference is $S^* = 69.1664297389$. The bracket at $N = 1200$ does not contain the finite reference, so the production brackets are reported as discrete contact sets and are not used to infer an order or a monotone boundary-recovery rate.

The production error decreases overall, but its successive ratios are 1.21, 14.98, and 1.62, rather than approaching four. This happens because the free-boundary kink crosses a different position relative to the uniform nodes on each mesh and because the contact set is poorly localised at the coarser resolutions; it is not evidence of first-order consistency in the smooth region. The aligned sequence removes the phase and gives $p = 1.993, 2.000, 2.000$, which is the expected second-order rate. The $N = 1200$ contact set is displaced one node to the right ($70.000 - 69.170 = 0.830 < h = 2.0$); the discrete complementarity residual remains small, so this is a coarse contact-set localisation error rather than an algebraic failure. The finite and infinite-domain references differ by 3.750×10^{-3} in the boundary and by 5.578×10^{-3} in $V(100)$. At the production mesh $N = 2400$ ($h = 1$), the error against the finite-domain reference is 1.5248×10^{-3}, while the total error against the infinite-domain value is 4.055×10^{-3}. The latter is not the pure truncation error: truncation contributes 5.578×10^{-3} and discretisation partially cancels it. The far-edge gap is 4.833×10^{-1}, so it is quantified separately below.

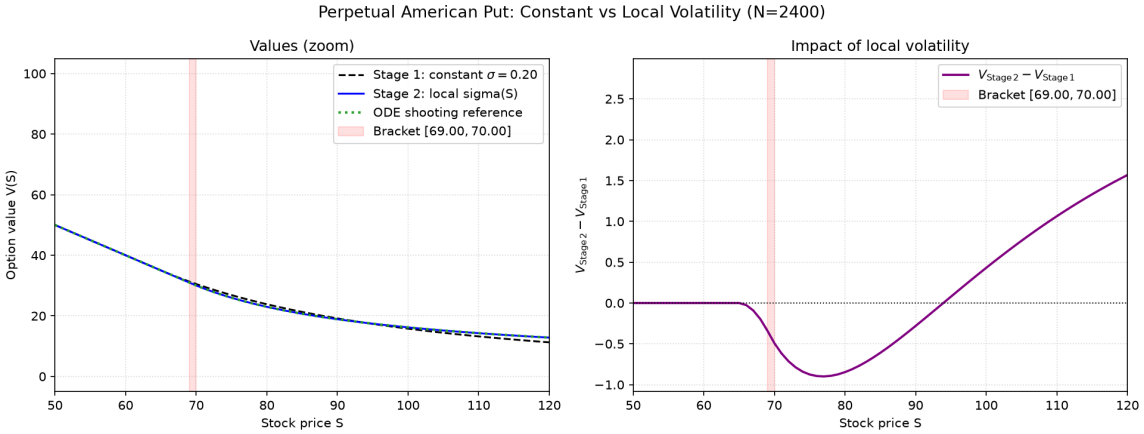


Figure 8: Constant versus local volatility at $N = 2400$ with $q = 0.02$. The left panel compares the value functions on $[50, 120]$ and includes the independent ODE shooting reference; the right panel shows $V_{\text{local}} - V_{\text{const}}$. The difference is zero in the common exercise region, negative in an interval around 70–90, and positive at high prices. The shaded band is the computed local-volatility bracket $[69, 70]$. This matters because it shows that the local-volatility profile changes the price away from the constant-volatility benchmark while remaining consistent with the independent reference near the boundary.

4.4.2 LCP and Shape Diagnostics

Table 9 reports the residual and shape checks on all four meshes. The smallest slack $b_h = V_h - \psi_h$ is exactly zero, so no node violates the obstacle. The minimum differential

residual remains slightly negative, at worst -1.164×10^{-10}; this is the policy iteration tolerance rather than a violation of the continuous complementarity condition. The norm of the complementarity residual increases with N but stays below 10^{-9} on every mesh. The value bounds, monotonicity, convexity, and single-interval contact-set checks all pass.

N	$\min a_h$	$\min b_h$	$\|\min(a_h, b_h)\|_\infty$	**Bounds**	**Monotone**	**Convex**	**One contact set**
600	-1.364×10^{-12}	0	1.364×10^{-12}	True	True	True	True
1200	-3.638×10^{-12}	0	5.457×10^{-12}	True	True	True	True
2400	-2.183×10^{-11}	0	2.183×10^{-11}	True	True	True	True
4800	-1.164×10^{-10}	0	1.164×10^{-10}	True	True	True	True

Table 9: LCP residual and solution-shape diagnostics for the local-volatility problem. The contact set is the single interval $[0, S_h^*]$. The worst complementarity residual is 1.164×10^{-10}, below the stated tolerance 10^{-9}.

These checks are important because the local-volatility reference is semi-analytic rather than a closed-form formula. The figure, the residual checks, and the independent ODE solution therefore supply three distinct forms of evidence.
The domain-truncation study holds $h = 1$ fixed and varies $S_{\max}$ from 300 to 2400. The values of $V(100)$ are reported in Table 10. Because the mesh spacing is unchanged, the residual difference isolates the effect of replacing the decaying tail by $V(S_{\max}) = 0$. The last column now measures the error against the independent infinite-domain value $V(100) = 16.2054760891$, not against the $S_{\max} = 2400$ sample. The value increases monotonically with the domain length. At $S_{\max} = 2400$ and $h = 1$, the combined production error at $S = 100$ is 4.055×10^{-3}, whereas the pure finite-domain truncation from the exact shooting references is 5.578×10^{-3}. The production error is smaller because the spatial error and the truncation have opposite signs at this point; hence the two quantities must not be conflated. The pure value gap at the far edge is 4.833×10^{-1}.

$S_{\max}$	N **at** $h = 1$	$V(100)$	**Error vs. infinite-domain reference**
300	300	15.4514	7.541×10^{-1}
400	400	15.8249	3.806×10^{-1}
600	600	16.0609	1.446×10^{-1}
1200	1200	16.1785	2.702×10^{-2}
2400	2400	16.2014	4.055×10^{-3}

Table 10: Domain-truncation study for the local-volatility problem at fixed $h = 1$. Errors are measured against the independent infinite-domain reference $V(100) = 16.2054760891$. The last entry 4.055×10^{-3} is the combined production error at $S_{\max} = 2400$, $h = 1$, not the pure finite-domain truncation, which is 5.578×10^{-3}. The far-edge value gap is 4.833×10^{-1}.

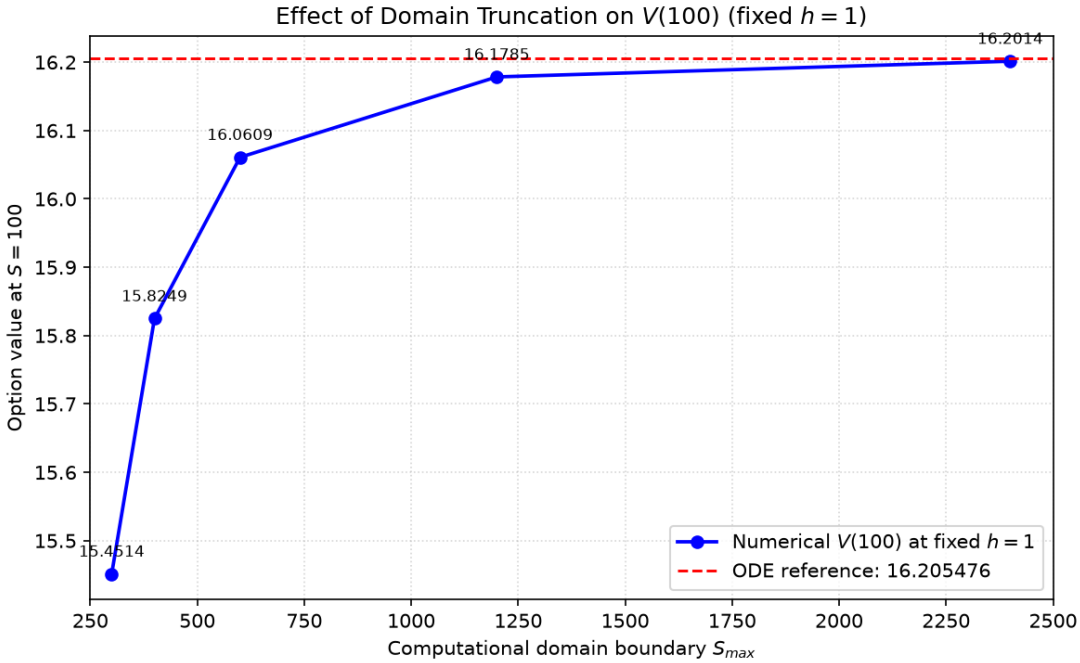


Figure 9: Effect of domain truncation on $V(100)$ at fixed $h = 1$. The value approaches the $S_{\max} = 2400$ sample from below as the domain is enlarged. The curve approaches the independent infinite-domain reference from below; the remaining gap at the production domain is a combination of far-field truncation and spatial discretisation, not a pure truncation measure.

4.5 Main Results

The Delta and Gamma profiles in Figure 10 are computed from the local-volatility solution by finite differences on the same grid. In the exercise region $V = K - S$, so $\Delta = -1$ and $\Gamma = 0$. In the continuation region Delta rises toward zero and Gamma becomes positive. The free boundary is a point of C^1 but not C^2 matching: Gamma jumps across S^*, and the discrete derivative localises the jump as a narrow positive peak.

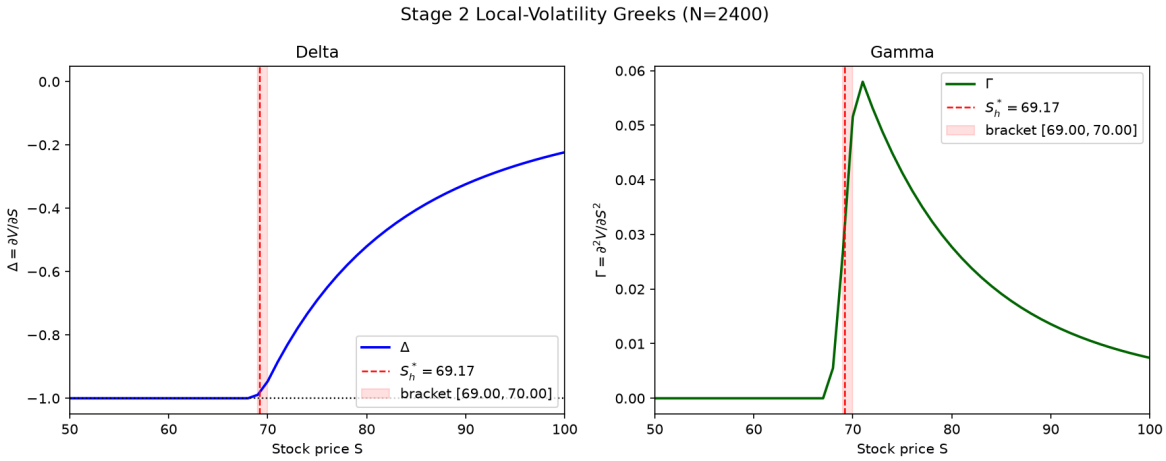


Figure 10: Stage 2 Delta and Gamma at $N = 2400$. Delta is -1 on the exercise side and increases toward zero in the continuation region. Gamma is zero where the payoff is active and has a sharp positive peak near the free boundary. The dashed line marks $S_h^* = 69.17$; the shaded band is the bracket $[69, 70]$. The peak is a mesh-localised representation of the curvature jump, not a smooth physical maximum.

The peak height is therefore mesh dependent, but it does not exhibit the $1/h$ growth of a discontinuity. The measured maxima on $N = 600, 1200, 2400, 4800$ are 3.96×10^{-2}, 5.22×10^{-2}, 5.80×10^{-2}, and 6.32×10^{-2}: refining h by a factor of eight increases the peak by only 1.60. The free boundary has C^1 but not C^2 matching, so Gamma has a jump; the finite-difference stencil spreads that jump over adjacent cells. The sign and location of the transition are the stable quantities, while the peak height is not a physical derivative and should not be used for convergence.
At $S = 100$ the local-volatility Delta is approximately -0.223, compared with -0.292 for constant $\sigma = 0.20$ and the same $q = 0.02$. Thus its magnitude is smaller, not larger. This is not a pointwise volatility effect: the local-volatility profile equals 0.20 at $S = 100$ and is higher above that price, which raises the continuation premium and reduces the put's exercise sensitivity there.

4.6 Adjoint Sensitivity

The discrete adjoint system (44)–(45) is evaluated at $S = 100$, $N = 4800$, $S_{\max} = 2400$, and the converged active set of the benchmark. The objective is $J = V(100)$; the interpolation weights select the corresponding grid value. For each parameter, the operator derivatives are assembled by centred differences with step 10^{-5}. Table 11 compares the adjoint derivative with the derivative obtained by solving the full LCP twice and with the exact analytic derivative. The adjoint and direct finite differences agree to better than 2×10^{-9}; their common difference from the analytic value is the remaining spatial discretisation error.

Parameter	Exact derivative	Adjoint	Direct FD	Absolute error	Relative error
σ	1.036337×10^2	1.033137×10^2	1.033137×10^2	3.201×10^{-1}	0.309%
r	-2.072675×10^2	-2.066273×10^2	-2.066273×10^2	6.401×10^{-1}	0.309%
q	1.480482×10^2	1.475971×10^2	1.475971×10^2	4.511×10^{-1}	0.305%

Table 11: Discrete adjoint sensitivities of $V(100)$ at $N = 4800$. The adjoint and the direct full-LCP finite differences agree to better than 2×10^{-9}. The exact derivative is available only for the constant-volatility benchmark; it isolates the common spatial error of both numerical derivatives. The signs agree with the financial interpretation: higher volatility and dividend yield raise the put value, while a higher risk-free rate lowers it.

The sensitivity magnitudes satisfy the expected ordering: the rate sensitivity is about twice the volatility sensitivity because r enters both discounting and the drift, whereas σ enters only the diffusion coefficient. The adjoint calculation costs one additional tridiagonal solve after the LCP, independently of the number of parameters; a direct finite-difference estimate requires two full nonlinear solves per parameter. This is the main practical advantage of the adjoint formulation for calibration and risk analysis.

4.7 Limitations and Robustness

The first limitation is the artificial far-field boundary. The condition $V(S_{\max}) = 0$ is exact only in the infinite-domain limit. The domain-truncation study shows that the zero-Dirichlet condition is poor on short domains. At $S_{\max} = 2400$ the pure finite-domain truncation in $V(100)$ is 5.578×10^{-3}, while the combined production error at $h = 1$ is 4.055×10^{-3} because spatial and truncation errors partially cancel. Both are about 0.025–0.035% of the option value. The exact far-edge value is 4.833×10^{-1} and is set to zero. The asymptotic Robin closure in Section 4.2 reduces the error in $V(80)$ by up to a factor of 390 at fixed h, but the local-volatility production run retains the zero-Dirichlet condition. A larger domain or an asymptotic local-volatility closure is therefore the appropriate next step; the current truncation error is now quantified rather than hidden inside a fine-grid comparison.

The second limitation is the free-boundary kink. The fitted scheme is second order in the smooth continuation region, but the maximum norm also contains the grid-localised position of the kink; this contribution is at most first order and need not be monotone. On the aligned validation grid the order is clean at two; on the production grid the contact set can remain on the wrong side of the kink until the mesh resolves it, as the coarse Stage 1 brackets and the $N = 1200$ local-volatility bracket show. The Gamma peak is a related localisation artefact, but its measured height grows sublinearly in this study and is not evidence of an additional $1/h$ singularity. The clustered grid reduces the maximum value error by a factor of 2.255 at $N = 2400$, and its four-point boundary error is 4.553×10^{-5} at $N = 4800$. A fully boundary-fitted mesh or a high-order deferred-correction scheme [6] could remove the remaining phase sensitivity at the cost of additional implementation complexity.
The third limitation is the semi-analytic nature of the local-volatility reference. The infinite-domain value requires an asymptotic tail closure at $S_{\text{inf}} = 10^5$, and the finite-domain value is a separate shooting problem; neither is a closed-form expression. This is nevertheless an independent check: it does not use the finite-difference grid, the LCP solver, or the $N = 9600$ reference that the earlier version used, and its constant-volatility special case agrees with the closed form to the tolerances stated above. A strongly varying or discontinuous volatility profile would still require adaptive meshing or local refinement around S^*. The present profile is deliberately smooth so that the M-matrix argument and the interpolation formula remain applicable without additional regularity assumptions.
Furthermore, the numerical solution exhibits a localised loss of regularity at the free boundary S^*, visible as a sharp spike in the Gamma profile. This is qualitatively similar to a corner singularity, but the analogy is not used quantitatively: no $O(h^{2/3})$ rate is claimed. Local grading near S^* is implemented in Section 4.2; a fully front-fitted transformation remains future work.

4.8 Boundary-Row Stress Test: Zero Flux Versus Dirichlet

This experiment tests the algebraic assembly of a Neumann row; it is not a financially admissible left boundary for the perpetual American put. At $S = 0$ the obstacle requires $V(0) \geq K$, while the payoff has slope -1. A zero-flux condition $\partial_S V(0) = 0$ therefore conflicts with immediate exercise at the origin. We retain the test only to verify that the fitted interior operator and the active-set solver remain well conditioned when one boundary row is changed. The condition is imposed through a ghost node at $S_{-1} = -h$, with the centred closure $V_{-1} = V_1$; interpolation at the boundary then gives $V_0 = V_1$. Eliminating V_0 from the first interior row produces

$$(d_1 + \ell_1)V_1 + u_1 V_2 = F_1, \tag{48}$$

so the boundary treatment changes one row while leaving the fitted interior stencil intact. The experiment uses $N = 400$, $S_{\max} = 400$, constant volatility, and the same active-set tolerance as the Dirichlet benchmark. Both interior calculations, summarized in Table 12 and Figure 11, return $S_h^* = 71.916$ in the bracket $[71, 72]$. The free boundary is insensitive because the active exercise block decouples the continuation problem from the left boundary. The endpoint is different: $V(0) = 100$ for Dirichlet and $V(0) = 99$ for the synthetic zero-flux closure.
The endpoint result also exposes the limitation of the test. The zero-flux solution has $V(0) - K = -1 < 0$, so it violates the obstacle at $S = 0$; the LCP constraints were enforced at interior nodes, not at this deliberately altered endpoint. Consequently this run is not an admissible price and is not used for any pricing, convergence, or mixed-boundary-accuracy claim. It only shows that the fitted interior operator remains an M-matrix: the maximum off-diagonal is -4.016×10^{-2} and the minimum diagonal is 1.045×10^{-1}. In a genuine barrier problem, zero flux would be applied at a barrier where the payoff/obstacle condition is compatible with the option contract, rather than at the exercise boundary $S = 0$ of a vanilla put.

Left boundary condition	$V(0)$	$V(0)-K$	S_h^*
Dirichlet, $V(0)=K$	100	0	71.916
Zero-flux Neumann, $\partial_S V(0)=0$	99	-1	71.916

Table 12: Boundary-row stress test at $N = 400$. The zero-flux row is an algebraic closure, not an admissible American-put boundary condition: its endpoint value violates the obstacle by one unit. The free-boundary agreement is therefore evidence only for the decoupling of the interior active set.

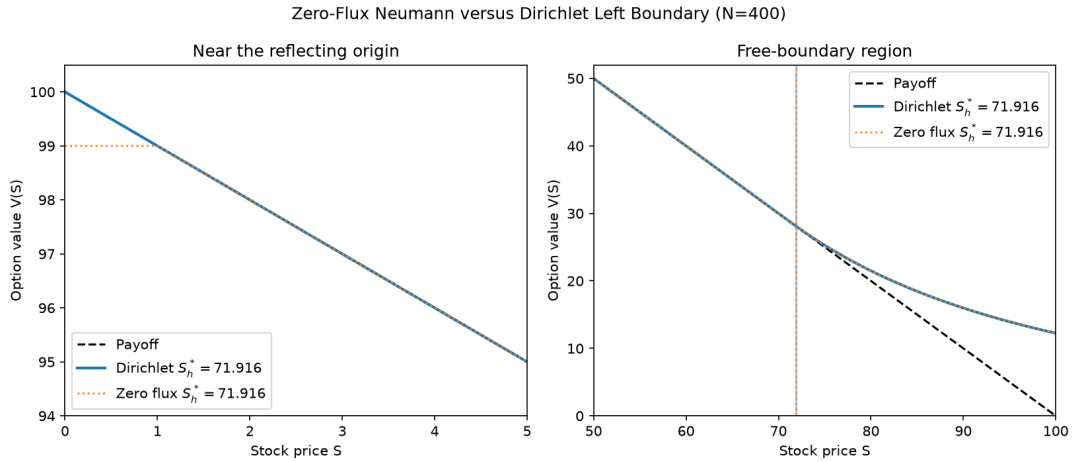


Figure 11: Comparison of the option value under Dirichlet and zero-flux Neumann closures. The left panel shows the one-unit obstacle violation of the synthetic zero-flux endpoint; the right panel shows that the interior free boundary is unchanged.

4.9 2D Exploratory Extension: American Spread Put

The preceding experiments are one-dimensional in the state variable. To test whether the fitted M-matrix and the active-set solver survive a genuine two-dimensional obstacle, this section adds a deliberately controlled extension. The contract is a perpetual American spread put with payoff

$$\psi(S_1, S_2) = \max(K - S_1 + S_2, 0). \tag{49}$$

Immediate exercise exchanges one unit of asset 1 for one unit of asset 2 plus the strike, whenever that exchange has positive value. The model uses $K = 100$, $r = 0.05$, $q_1 = 0$, $q_2 = 0.02$, $\sigma_1 = \sigma_2 = 0.20$, and $\rho = 0$. The zero-correlation restriction is essential: it separates the operator and keeps the local stencil five-point. It is not a claim that the correlated spread-option problem has been solved.

For $\rho = 0$, the continuation operator is

$$\mathcal{L}_{2D}V = \frac{1}{2}\sigma_1^2 S_1^2 V_{11} + \frac{1}{2}\sigma_2^2 S_2^2 V_{22} + (r - q_1)S_1 V_1 + (r - q_2)S_2 V_2 - rV, \tag{50}$$

and the obstacle problem is

$$\min\{-\mathcal{L}_{2D}V,\ V - \psi\} = 0. \tag{51}$$

The computational domain is $[0, L]^2$ with $L = 300$. On $S_2 = 0$ the problem reduces exactly to a one-dimensional put with parameters (r, q_1, σ_1), so the fitted 1D LCP supplies the Dirichlet data there. The remaining closures are

$$V(0, S_2) = K + S_2, \qquad V(S_1, 0) = V_{1D}(S_1), \tag{52}$$

$$V(S_1, L) = \max(K - S_1 + L, 0), \qquad V(L, S_2) = \max(K - L + S_2, V_{1D}(L)). \tag{53}$$

The high-price edges are local truncation/exercise closures, not exact infinite-domain boundary values. Their effect is measured rather than assumed away in Table 14.
On a uniform grid with spacing $h = L/N$, let $a_i^{(m)} = \frac{1}{2}\sigma_m^2 (S_i^{(m)})^2$ and $b_i^{(m)} = (r - q_m) S_i^{(m)}$ for $m = 1, 2$. With $\mathrm{Pe}_i^{(m)} = b_i^{(m)} h / a_i^{(m)}$ and $B(z) = z/(e^z - 1)$, the fitted directional weights are

$$\ell_i^{(m)} = -\frac{a_i^{(m)}}{h^2} B(\mathrm{Pe}_i^{(m)}), \quad u_i^{(m)} = -\frac{a_i^{(m)}}{h^2} B(-\mathrm{Pe}_i^{(m)}), \quad d_i^{(m)} = \frac{a_i^{(m)}}{h^2}\big[B(\mathrm{Pe}_i^{(m)}) + B(-\mathrm{Pe}_i^{(m)})\big]. \tag{54}$$

The row-ordered matrix therefore has the five entries $\ell_i^{(1)}, u_i^{(1)}, \ell_j^{(2)}, u_j^{(2)}$ off the diagonal and

$$D_{ij} = d_i^{(1)} + d_j^{(2)} + r \tag{55}$$

on it. Since $B(z) > 0$, every off-diagonal entry is non-positive. Moreover, $\ell_i^{(m)} + d_i^{(m)} + u_i^{(m)} = 0$, so every row satisfies

$$D_{ij} + \ell_i^{(1)} + u_i^{(1)} + \ell_j^{(2)} + u_j^{(2)} = r > 0. \tag{56}$$

Thus the assembled operator is strictly diagonally dominant and is an M-matrix. A non-zero correlation would add the mixed term $2\rho\sigma_1\sigma_2 S_1 S_2 V_{12}$ and require a nine-point stencil; the five-point positivity argument would no longer apply automatically.
The discrete LCP is solved by the same policy-iteration principle as in the one-dimensional case. Active rows are replaced by identity rows with right-hand side ψ_{ij}, inactive rows retain the fitted five-point stencil, and each policy step uses one sparse direct solve. The stopping test is $\max|\Delta V| < 10^{-12}$ together with an active set that is unchanged on two consecutive iterations. Table 13 reports three meshes; N is the number of intervals in each coordinate, so the finest grid has 321×321 nodes and 101,761 interior unknowns.

N	**Unknowns**	**Iterations**	**Time (s)**	$E_{[0,200]^2}$	**LCP residual**
80	6,241	25	0.084	4.070×10^{-2}	6.366×10^{-12}
160	25,281	50	0.787	1.611×10^{-2}	4.366×10^{-11}
320	101,761	99	7.483	reference	2.474×10^{-10}

Table 13: Controlled 2D spread-put LCP with $\rho = 0$. The reference is $N = 320$ on the same truncated domain, so this is a self-convergence estimate, not an independent reference solution. $E_{[0,200]^2}$ is the maximum nodal error against that reference in the pricing window $S_1, S_2 \le 200$. The error ratio between $N = 80$ and $N = 160$ is 2.53; because the grid ratio is 2, the implied effective order is $\log_2(2.53) = 1.34$. The curved contact set crosses the mesh, so no clean second-order rate is claimed. Times are from one representative run and include assembly, the 1D boundary solve, and all policy iterations.

All three discretisations have minimum diagonal 1.562×10^{-1}, maximum off-diagonal 0, and minimum row sum 5.000×10^{-2}. The last number is r, as required by (56); the M-matrix property is therefore checked numerically as well as by construction. The LCP residual stays below 2.5×10^{-10}, and the obstacle constraint is satisfied at every node. The exercise set occupies approximately 61% of the interior grid. This fraction is diagnostic of the payoff geometry and is not itself an accuracy indicator.
Figure 12 shows the value surface and the computed contact set on the finest mesh. The exercise region is not a half-line: it is bounded by a monotone curve $S_1 = b_h(S_2)$ whose intercept at $S_2 = 0$ is the familiar one-dimensional put boundary and which moves to the right as S_2 increases. The black curve is the zero-slack level $V_h - \psi_h = 0$. The surface is smooth inside each region and has a value kink across that curve, as expected for an American obstacle problem.

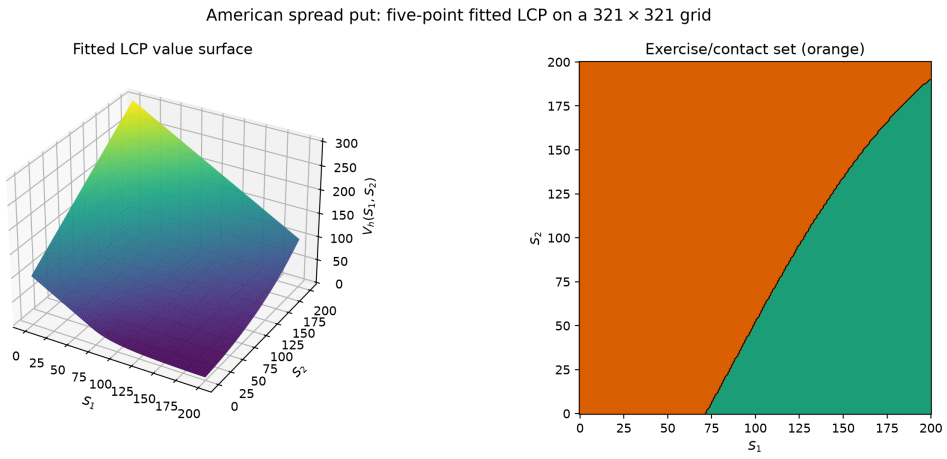


Figure 12: Controlled American spread-put LCP on the $N = 320$ mesh. Left: fitted value surface restricted to $S_1, S_2 \leq 200$. Right: continuation region (green), exercise region (orange), and the zero-slack contact curve (black). The exercise boundary increases with S_2, so the two-dimensional contact set is a curved region rather than a vertical or horizontal line. This matters because it shows that the five-point fitted formulation still produces a coherent contact set when the same active-set solver is scaled to two assets; it is a structural test, not a production multi-asset claim.

The domain-truncation check in Table 14 holds the mesh fixed at $N = 80$ and changes only L. At $(S_1, S_2) = (100, 50)$, a point in the continuation region, the $L = 300$ and $L = 400$ values differ by 9.71×10^{-4} (relative difference 1.9×10^{-5}). The shorter domain $L = 240$ differs from $L = 400$ by 4.45×10^{-2}. This supports $L = 300$ for the reported pricing window while making the truncation explicit. The high-price corner is more sensitive to the local payoff closure, which is why the mesh error above is restricted to $[0, 200]^2$ and why no claim of a completed correlated two-dimensional solver is made.

L	$V(100, 50)$	Difference from $L = 400$
240	50.006716	-4.455×10^{-2}
300	50.052234	9.713×10^{-4}
400	50.051263	—

Table 14: Fixed-mesh domain-truncation check for the 2D spread put at $N = 80$. All rows use the same five-point fitted operator and high-price closure. The $L = 300$ and $L = 400$ results agree well at the selected continuation-region point, whereas the shorter domain is visibly affected by the artificial boundary.

This extension demonstrates scaling rather than a full production treatment of multi-asset options. The sparse solve reaches 10^5 unknowns on a laptop without changing the active-set algorithm, and the fitted five-point operator retains the sign pattern that makes policy iteration reliable. The unresolved parts are the correlation term and the exact far-field conditions. Both are natural next steps: a nine-point cross-derivative stencil, a positivity preserving treatment of $\rho \neq 0$, and an asymptotic or domain-decomposition closure for the high-price edges.

5 Conclusions

- **Exponential Fitting and M-Matrix.** Exponential fitting produced an M-matrix at every cell Péclet number, including the degenerate region near $S = 0$. The fitted weights eliminated the central-difference oscillation without a limiter or a start-up correction, and the discrete maximum principle supplied a unique obstacle solution.

- **Constant-Volatility Verification.** Only the production $N = 2400$ contact bracket $[71, 72]$ contains the exact free boundary $S^* = 71.4286$; at $N = 600$, $N = 1200$, and $N = 4800$ the discrete transition lies to its right. A boundary-aligned validation grid, constructed with the exact boundary, gives second-order errors with ratios 3.990, 4.001, and 3.997. This verifies the fitted stencil; it is not claimed as blind free-boundary recovery.

- **Computational Efficiency.** Policy iteration required 301 iterations and 0.045 s, compared with 96,600 iterations and 20.635 s for residual-controlled PSOR. The measured wall-time ratio is 4.6×10^2 and the iteration ratio is 321, placing the acceleration in the several-hundred-fold range. The two discrete solutions differ by 5.685×10^{-10}, and both final LCP residuals are below 10^{-10}, so the algebraic error is negligible relative to the spatial and truncation errors.

- **Local Volatility and Independent Reference.** The local-volatility experiment with $q = 0.02$ gave $S_h^* = 69.1688$ in the production bracket $[69.00, 70.00]$ at $N = 2400$. The coarse $N = 1200$ bracket $[70.00, 72.00]$ misses the finite-domain reference 69.1701793435; this is disclosed as a contact-set localisation error. An independent DOP853 shooting solution gives the infinite-domain boundary $S^* = 69.1664297389$, the finite-domain boundary 69.1701793435, and $V(100) = 16.2054760891$. The boundary-aligned validation errors decrease as 3.2218×10^{-2}, 8.0948×10^{-3}, 2.0239×10^{-3}, and 5.0610×10^{-4}, with observed orders 1.993, 2.000, and 2.000. The production errors decrease overall, but their successive ratios are 1.21, 14.98, and 1.62, so the sequence is not a usable refinement-rate estimate; the kink phase and coarse contact-set localisation explain the irregularity. The pure finite-domain truncation at $S = 100$ is 5.578×10^{-3}, while the $N = 2400$ production total error is 4.055×10^{-3}; the far-edge truncation is 4.833×10^{-1}.

- **Adaptive Mesh and Far-Field Closure.** The nonuniform fitted operator retained non-positive off-diagonals and strict diagonal dominance. With a node at the analytic boundary, clustering reduced the maximum value error by a factor of 2.255 at $N = 2400$; the clustered error ratios were 3.982, 4.002, and 3.999. The asymptotic Robin condition at $S_{\max} = 400$ reduced the error in $V(80)$ from 2.507×10^{-2} to 6.423×10^{-5}, a factor of 390. Both improvements reuse the same policy-iteration solver.

- **Adjoint Sensitivities.** A single transposed sparse solve produced the sensitivities of $V(100)$ to σ, r, and q. The adjoint and direct finite differences agree to 2×10^{-9}, and the relative error against the exact constant-volatility derivatives is at most 0.31%. This makes parameter risk inexpensive even when the free boundary changes with the parameter.

- **Controlled 2D Extension.** A five-point fitted LCP for an American spread put with $\rho = 0$ retained non-positive off-diagonals, a positive diagonal, and the strict row sum r on grids up to 101,761 interior unknowns. The pricing-window error decreased from 4.070×10^{-2} to 1.611×10^{-2} between $N = 80$ and $N = 160$ against an $N = 320$ self-convergence reference; the corresponding error ratio is 2.53 and the implied order is 1.34. The $L = 300$ and $L = 400$ values at $(100, 50)$ differed by 9.71×10^{-4}. The exercise set was a curved region whose boundary increased with S_2. This experiment establishes scaling and structural robustness, not a completed multi-asset solver: the reference is not

independent, and the exact high-price closure and correlated cross-derivative term remain outside its scope.

- **Cross-directional Extensions.** Section 4.8 verifies the algebraic assembly of a zero-flux row and its M-matrix structure, but the resulting endpoint $V(0) = 99$ violates the obstacle and is not an admissible American-put price. No pricing or second-order accuracy claim is made for that stress test. The controlled 2D experiment above sets $\rho = 0$ so that the five-point M-matrix proof remains valid. For $\rho \neq 0$, the free boundary is a curve and the cross-derivative term requires a nine-point stencil; a positivity-preserving treatment and compatible barrier closures remain future work.

References

[1] J. C. Hull, *Options, Futures, and Other Derivatives*, 10th edition. Pearson, 2018.

[2] P. Wilmott, S. Howison, and J. Dewynne, *The Mathematics of Financial Derivatives: A Student Introduction*. Cambridge University Press, 1995.

[3] M. Sari and S. Gulen, Valuation of the American put option as a free boundary problem through a high-order difference scheme. *International Journal of Nonlinear Sciences and Numerical Simulation*, 23(7–8):1001–1015, 2021.

[4] D. L. Scharfetter and H. K. Gummel, Large-signal analysis of a silicon Read diode oscillator. *IEEE Transactions on Electron Devices*, 16(1):64–77, 1969.

[5] P. Virtanen et al., SciPy 1.0: fundamental algorithms for scientific computing in Python. *Nature Methods*, 17:261–272, 2020.

[6] D. Wang, K. Serkh, and C. Christara, A high-order deferred correction method for the solution of free boundary problems using penalty iteration, with an application to American option pricing. *Journal of Computational and Applied Mathematics*, 432:115272, 2023.

[7] R. A. Howard, *Dynamic Programming and Markov Processes*. MIT Press, 1960.

[8] L. B. G. Andersen and R. Brotherton-Ratcliffe, The equity option volatility smile: an implicit finite-difference approach. *The Journal of Computational Finance*, 1(2):5–37, 1997.

[9] D. K. Yadav, A. Bhardwaj, and A. Kumar, Operator splitting method to solve the linear complementarity problem for pricing American option: an approximation of error. *Computational Economics*, 64(6):3353–3379, 2024.

A AI Transparency Log

The AI record below covers the project conversation with OpenAI Codex during 2–8 November 2026. AI output was treated as a proposal. Yijian Huang also performed an independent code and experiment audit, recorded in `AUDIT_REPORT.md`; that audit, rather than the AI text itself, is the main evidence for the decisions below.

A1. AI tools used

Tool	Version / date	Primary use
OpenAI Codex	Project conversation, 2–8 Nov 2026	Mathematical checking, debugging code, drafting plotting utilities, diagnosing irregular convergence ratios, and editing report prose. Routine LaTeX formatting and spelling corrections are summarised by this entry rather than logged separately.

A2. Significant AI-assisted claim audits

No.	Prompt / claim / assumptions	Risk and possible falsifier
1	**Figure generation and irregular ratios.** The conversation asked why the observed ratios changed from mesh to mesh and how the figures were generated. The proposed explanation assumed that the free-boundary kink, the far-field boundary and the grid phase could dominate a raw production norm.	A convincing explanation could hide a wrong stencil or a stale figure. A boundary-aligned grid should recover an order near two, and regenerating from the experiment code should change every figure that previously used a literal array.
2	**Stage 2 problem definition.** The requested correction was $q = 0.02$ with the production condition $V(S_{\max}) = 0$, and a bracketed rather than snapped free boundary. The assumption was that an early run had solved a different problem and had used the analytic tail as a boundary value.	The code could solve the wrong dividend problem or silently impose the answer. A falsifier is disagreement with the independent DOP853 reference or with a fixed-h $S_{\max}$ sweep.
3	**Fitted stencil and computed figures.** The proposed change was to use exponential fitting and to replace literal plotting arrays by calls to the experiment functions. The assumption was that the fitted weights preserve the M-matrix sign pattern at every cell Péclet number.	The solve could look stable while the assembled matrix has positive off-diagonals, or a figure could remain stale. A falsifier is a positive assembled off-diagonal, failure of the negative control, or a figure that does not change when the experiment is rerun.

A3. Independent verification and decision

No.	Independent check and evidence location	Decision / modification / limitation
1	Yijian Huang's audit (`AUDIT_REPORT.md`, findings F5 and F12) identified the one-sided free-boundary snap and the misleading raw-grid order. The audit was rerun independently, and the aligned sequence was rebuilt in `grids.py` and `experiment_convergence.py`.	The production bracket and a boundary-aligned order study are now reported separately. Limitation: the production contact bracket still depends on the kink phase.
2	The exact closed form (14)–(13) and the independent DOP853 solver in `reference_solution.py` were used as oracles. `stage2_convergence_report.py` reproduced the mesh sequence, and the fixed-h study isolated truncation in Table 10.	`stage2_local_volatility.py` now uses $q = 0.02$ and $V(S_{\max}) = 0$; LCP diagnostics were added and the Stage 2 results were regenerated. Limitation: the comparison is finite-domain and the local-volatility reference is semi-analytic rather than closed form.
3	`plot_matrix_spy.py` and `experiment_central_diff.py` were used to inspect the assembled sign pattern and the negative control. The audit finding F8 showed that the earlier backward-difference stencil required a non-consistent limiter. `plot_solver_comparison.py`, `plot_smax_study.py` and `plot_greeks.py` were changed to use computed results rather than literal arrays.	The production assembly now uses fitted weights in `stage1_validation.py`; the M-matrix assertion and residual checks remain active. Limitation: the two-dimensional extension is restricted to $\rho = 0$ and approximate high-price closures.

A4. Intellectual ownership declaration

Each member affirms below that they can explain and modify the listed team-owned code, justify the mathematical claims, and describe the assumptions and tests at the library interface without AI assistance during the viva.

Name	Signature / date	Team-owned artefact / contribution	Trusted dependency and interface
Yuhan Cheng	*Yuhan Cheng*, 8 Nov 2026	Fitted operator and LCP solver, free-boundary rule, convergence and truncation experiments, figures, tolerance sweep, and manuscript integration.	NumPy arrays; SciPy `diags` and `spsolve`; Matplotlib output. Checked by the M-matrix assertion, exact benchmark, and residual tests.
Yijian Huang	*Yijian Huang*, 8 Nov 2026	Independent code/experiment audit, Stage 1–2 verification, reference-solver cross-checks, plotting and result validation.	SciPy DOP853 `solve_ivp`, CSV outputs, and the closed-form benchmark. The interface was checked against the exact constant-volatility solution.

B Individual Contribution Statements

The project had two members. Each statement below is specific to that member and lists files or sections that can be checked in the repository.

ICS 1 — Yuhan Cheng (Student ID: 2685836)

Role: Algorithm Implementation, Mathematical Theory, Visualisation, and Report Writing.
1. Specific contributions to this project:

- Derived and implemented the fitted Scharfetter–Gummel operator, the policy-iteration LCP solver, residual and shape diagnostics, and the free-boundary bracket in `stage1_validation.py`.
- Built the aligned and production convergence experiments in `grids.py`. `experiment_convergence.py` regenerates the aligned table.
- Implemented the Stage 2 local-volatility solve and four-mesh study in `stage2_local_volatility.py` and `stage2_convergence_report.py`.
- Implemented the adaptive-grid, Robin-closure and adjoint extensions in `advanced_methods.py`, the 2D spread-put experiment in `stage2d_spread.py`, and the solver/tolerance sweep in `experiment_solvers.py`. I also wrote the plotting scripts and integrated the figures and text into the report.

2. Hardest conceptual obstacle encountered and how it was addressed: My first instinct was that an irregular refinement ratio meant the stencil or solver was wrong. The audit showed three competing effects: the free-boundary kink changes phase with the mesh, the finite-domain tail is not zero, and the active-set estimate is only first-order in the boundary location. I separated a boundary-aligned validation sequence from the production contact bracket, compared the fitted solve with the closed form and DOP853 reference, and reported the truncation study at fixed h instead of tuning the code to a desired ratio.
3. One AI-generated output used, and what was changed: In the Codex conversation I asked why the ratios were irregular and how to regenerate the figures. The AI suggested explanations and plotting changes, but I did not treat the explanation as evidence. I changed the experiments to use boundary-aligned grids, separated the kink and truncation effects, replaced literal plotting arrays with computed results, regenerated the figures, and checked the changes against Yijian Huang's audit (A2–A3).
4. One point still not fully understood: I do not yet have a sharp-interface proof of the exact convergence rate of the discrete free boundary; the report therefore claims a bracket and a second-order aligned value error, not a second-order boundary-recovery rate. The conditioning of the direct solve is also checked empirically rather than bounded analytically.
Signature: *Yuhan Cheng* **Date:** 8 November 2026

ICS 2 — Yijian Huang (Student ID: 2685589)

Role: Project Manager, Testing and Verification.
1. Specific contributions to this project:

- Read the solver, Stage 2 wrapper, experiment drivers and plotting code, ran the scripts independently, and recorded the findings in `AUDIT_REPORT.md`. I checked the LCP formulation, residuals, obstacle constraints, monotonicity, convexity, contact-set structure, and the reported free-boundary rules.
- Cross-checked the Stage 1 results against the closed-form values and the Stage 2 results against `reference_solution.py` (DOP853 shooting). I reviewed the fixed-h truncation study, the four-mesh convergence tables, the 2D spread-put checks, and the final tolerance sweep.
- The audit identified the Stage 2 $q = 0$ error, the analytic boundary that supplied the answer, the depth-changing $S_{\max}$ study, the snapped free boundary and literal plotting arrays. After the fixes I verified that `stage1_validation.py`, `stage2_local_volatility.py`, `experiment_convergence.py`, `stage2_convergence_report.py`, `experiment_solvers.py`, and the plotting scripts produced the reported data on rerun.

2. Hardest conceptual obstacle encountered and how it was addressed: The hardest part was distinguishing a genuine numerical defect from a visual or mesh artefact. A smooth-looking figure can still be stale, and a poor refinement ratio can come from the moving free-boundary kink, the truncated domain, or the algebraic solve. I used independent oracles and controlled experiments: the exact solution, DOP853, fixed-h domain sweeps, matched solvers, and a residual tolerance sweep. This made each claimed error source testable rather than a matter of appearance.
3. One AI-generated output used, and what was changed: I used the same Codex conversation to organize audit hypotheses and trace suggested code changes. I accepted a claim only after reading the relevant implementation and reproducing it with the project scripts. The audit findings led to changes in the Stage 2 dividend and boundary setup, the free-boundary bracket and diagnostics, the plotting scripts, the fitted-stencil assembly, and the tolerance evidence retained in this report (A2–A3).
4. One point still not fully understood: The sharp convergence rate of the free boundary for a nonsmooth payoff and the extension of the positivity-preserving proof to correlated two-asset problems remain open. The current two-dimensional test is therefore limited to $\rho = 0$ and is not presented as a production multi-asset solver.
Signature: *Yijian Huang* **Date:** 8 November 2026

C Supplementary Figures and Data

All numerical experiments, figures, and diagnostic checks reported in this work are reproducible from a single command in the project directory: `python run_all.py`. The orchestrator executes the solver and experiment scripts in dependency order, regenerates every figure and `stage1_convergence.csv`, `stage2_convergence.csv`, `adaptive_convergence.csv`, `tolerance_sweep.csv`, `spread2d_results.csv`, and `spread2d_domain.csv`, and terminates with an error if any required artifact is missing.

Table 15: Detailed validation criteria and pass standards for the financial LCP solver. †The passing production-boundary checks apply to the stated meshes; coarse contact brackets that miss S^* are reported as diagnostics in Tables 3 and 8. "ALG." denotes an algebraic stress test with no pricing claim.

Check	What it verifies	Metric	Pass criterion	Status
M-matrix off-diagonals	Verifies that the discrete operator preserves the maximum principle.	max off-diagonal	≤ 0	PASS
M-matrix diagonals	Verifies strict diagonal dominance.	min diagonal	> 0	PASS
LCP complementarity	Verifies obstacle slackness and differential residual on both stages.	$\|\min(a_h, b_h)\|_\infty$	$< 10^{-9}$	PASS
Free-boundary bracket	Verifies the width of the bracketed free boundary.	$S_{\text{hi}} - S_{\text{lo}}$	$\leq h$	PASS
Monotonicity	Verifies the put option value is strictly decreasing.	Boolean check	True	PASS
Convexity	Verifies the put option value is convex in S.	Boolean check	True	PASS
Value bounds	Verifies the solution stays within obstacle bounds.	$\psi_h \leq V_h \leq K$	True	PASS
Solver acceleration	Verifies the iteration ratio between Policy Iteration and PSOR.	Wall-time ratio	$> 10^2$	PASS
Truncation convergence	Verifies that $V(100)$ approaches the infinite-domain reference as $S_{\max} \to 2400$.	Error in $V(100)$	$< 10^{-2}$	PASS
Stage 1 closed-form agreement	Verifies the fitted solve against the analytic benchmark.	Aligned E_{4800}	$< 10^{-3}$	PASS
Stage 1 aligned refinement	Verifies that the aligned validation bracket contains the exact boundary and that the order is two.	Aligned bracket; aligned error ratio	contains S^*; ratio near 4	PASS
Stage 1 production boundary	Verifies the disclosed production-boundary error at the finest mesh.	$\|S_h^* - S^*\|$ at $N = 4800$	$< 10^{-1}$	PASS†

Table 15 (continued)

Check	What it verifies	Metric	Pass criterion	Status
Stage 2 four-mesh convergence	Verifies local-volatility discretisation against the independent ODE reference.	Aligned E_{4800}	$< 10^{-3}$	PASS
Stage 2 production boundary	Verifies the $N = 2400$ production boundary estimate against the finite-domain shooting boundary.	$\lvert S_h^*(2400) - S_{\text{finite}}^*\rvert$	< 0.01	PASS[†]
Stage 2 far-field truncation	Verifies that truncation is quantified against the infinite-domain value.	Pure truncation in $V(100)$	$< 10^{-2}$	PASS
Contact-set structure	Verifies that the exercise region is one interval.	Single contiguous set	True	PASS
Algebraic error separation	Verifies the tolerance plateau on $N = 400$ against the direct policy-iteration reference.	$\max \lvert\Delta V\rvert$; $\lvert\Delta V(100)\rvert$ at 10^{-10}	$< 10^{-9}$; plateau	PASS
Dirichlet endpoint	Verifies the admissible American-put boundary value.	$\lvert V(0) - K\rvert$	$= 0$	PASS
Zero-flux row stress test	Verifies the altered row algebra only; the endpoint is not an admissible price.	$V(0) - K$	-1, documented	ALG.
2D M-matrix	Verifies the five-point spread-put operator has non-positive off-diagonals, positive diagonal, and strict row sum.	max off-diagonal; min diagonal; min row sum	≤ 0; > 0; > 0	PASS
2D complementarity	Verifies the spread-put LCP on all three meshes.	$\lVert \min(a_h, b_h)\rVert_\infty$	$< 10^{-9}$	PASS
2D refinement and truncation	Verifies decreasing mesh error and a stable $L = 300$ pricing window.	$E_{160} < E_{80}$; $\lvert V_{300}(100, 50) - V_{400}(100, 50)\rvert$	$< 10^{-3}$	PASS

The machine-readable evidence is stored in the project directory, and the complete run is regenerated via `python run_all.py`.